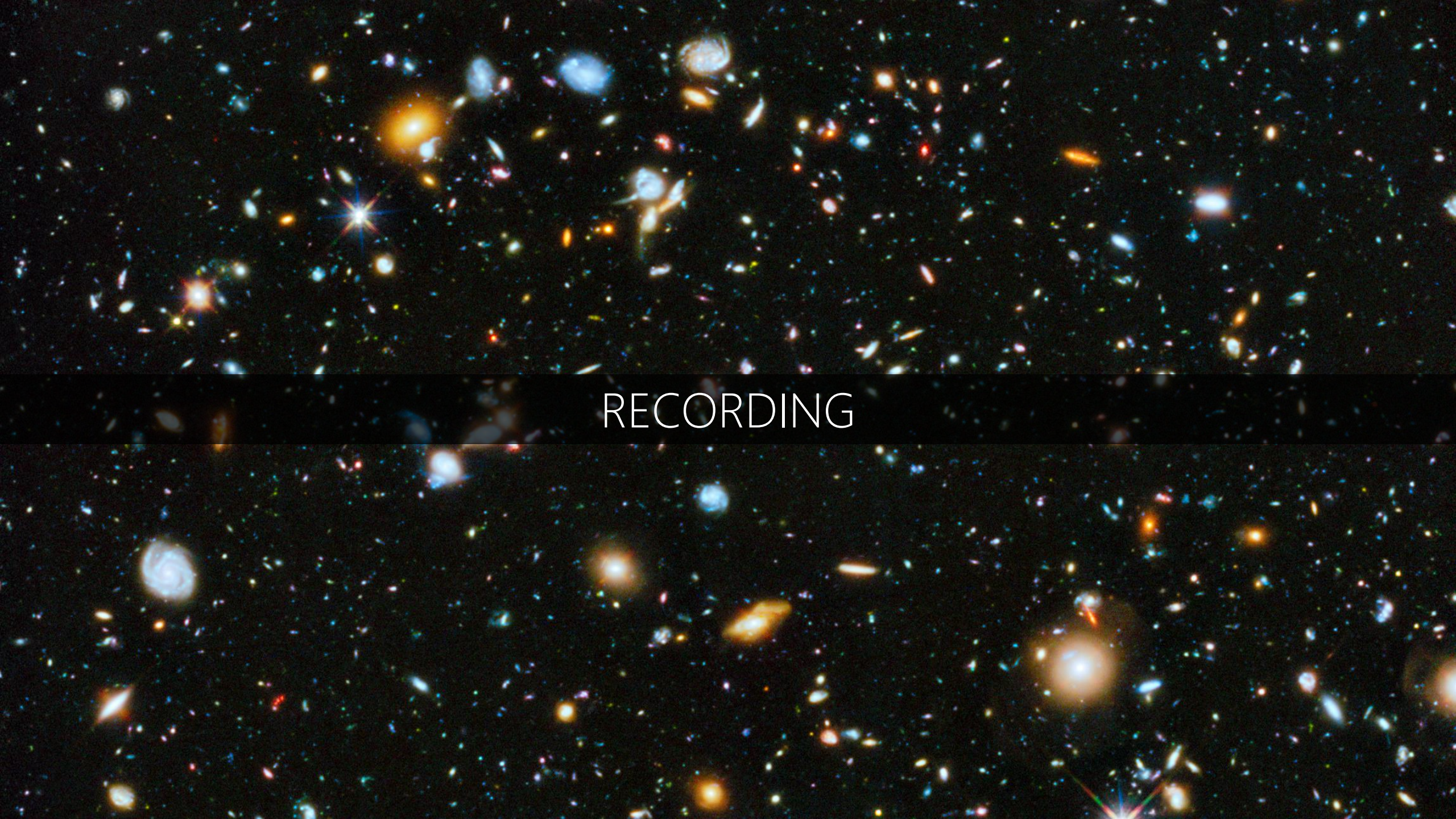


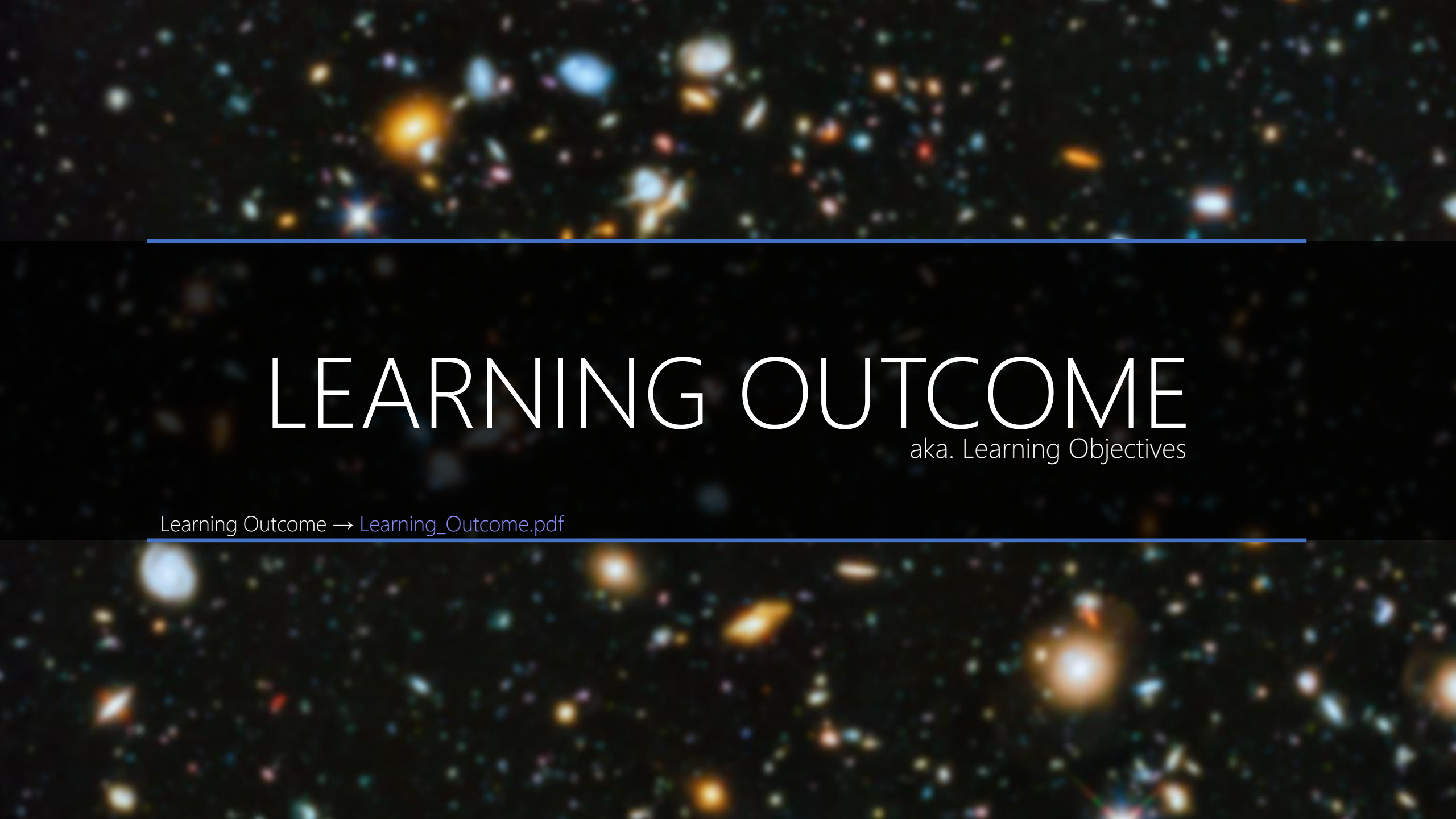


RECORDING

HEADS UP

The background of the entire image is a deep space photograph showing a vast field of galaxies. These galaxies appear as bright, colorful spots in various shapes and sizes, including spirals, ellipticals, and irregular forms, scattered across a dark, star-filled sky. Two thin, horizontal blue lines are positioned above and below the central text, spanning most of the image width.

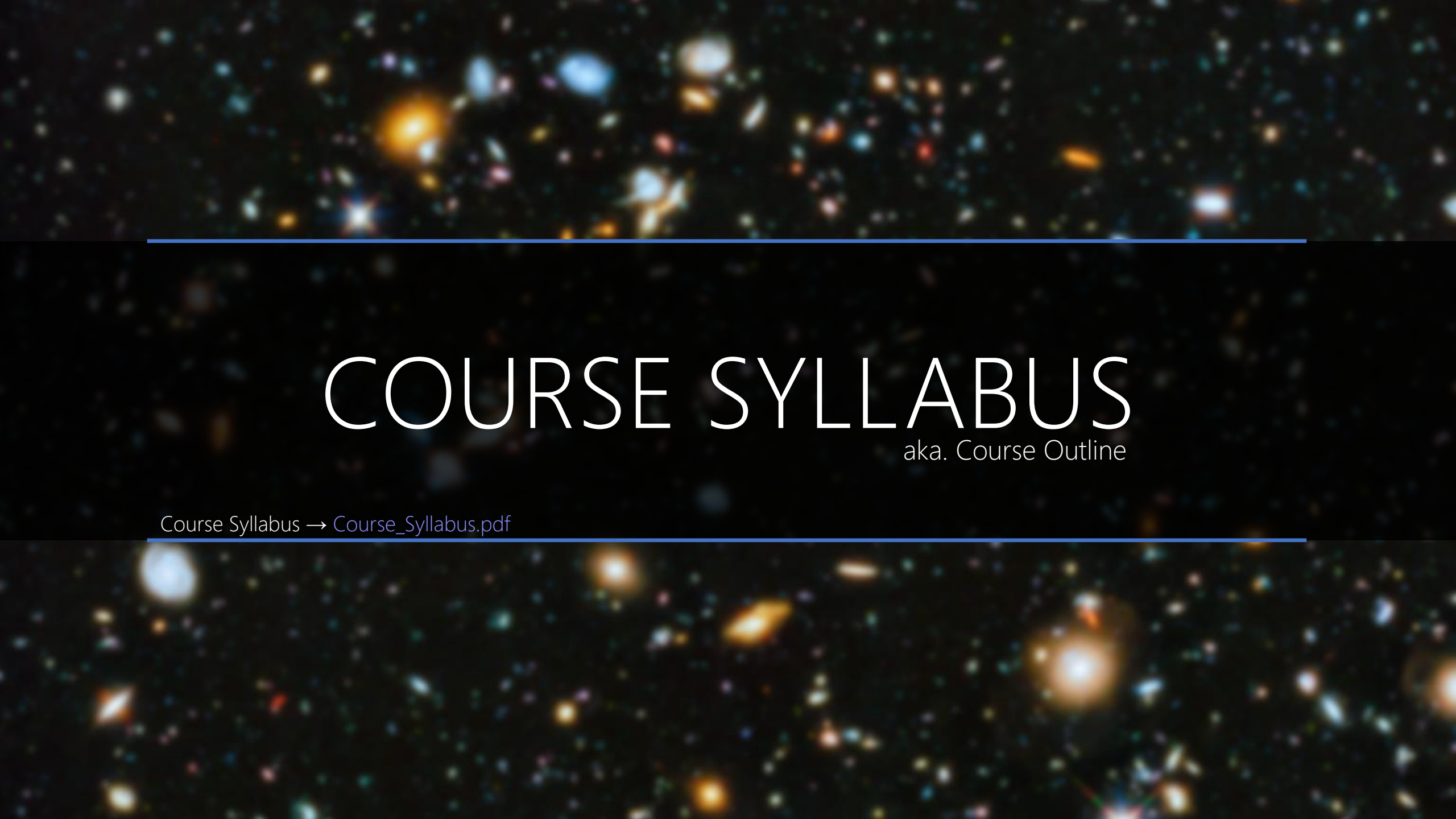
BRIGHTSPACE TOUR



LEARNING OUTCOME

aka. Learning Objectives

Learning Outcome → [Learning_Outcome.pdf](#)

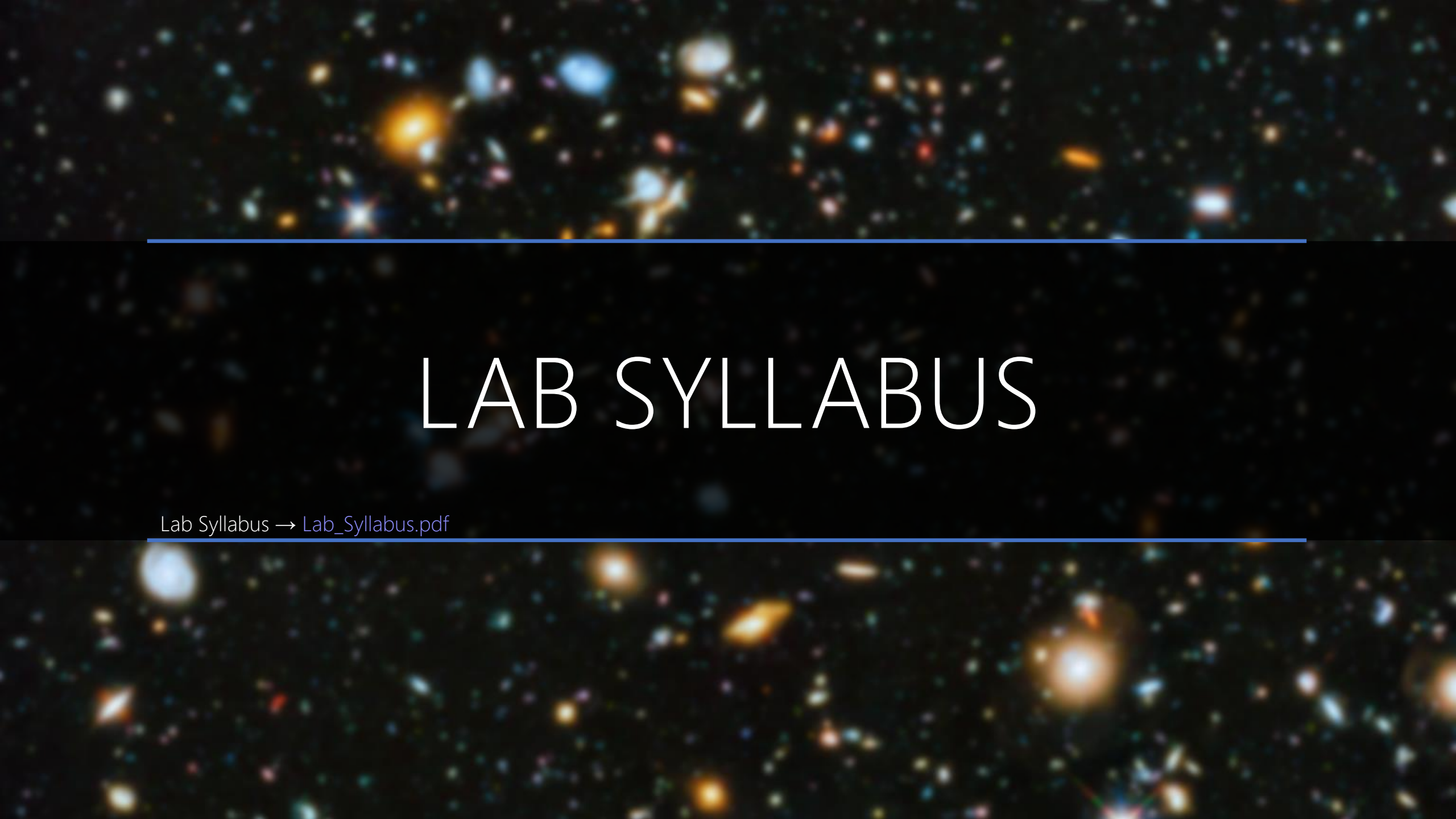
A cosmic background image featuring a dense field of galaxies in various colors (yellow, orange, blue, red) against a black space. A horizontal blue line is positioned above the main title.

COURSE SYLLABUS

aka. Course Outline

Course Syllabus → [Course_Syllabus.pdf](#)

A cosmic background image featuring a dense field of galaxies in various colors (yellow, orange, blue, red) against a black space. A horizontal blue line is positioned below the text.

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, fuzzy blobs and others as more distinct, elongated structures. A thin blue horizontal line is positioned above the title.

LAB SYLLABUS

Lab Syllabus → [Lab_Syllabus.pdf](#)

A continuation of the cosmic background image from the top, showing more galaxies. A thin blue horizontal line is positioned below the text.



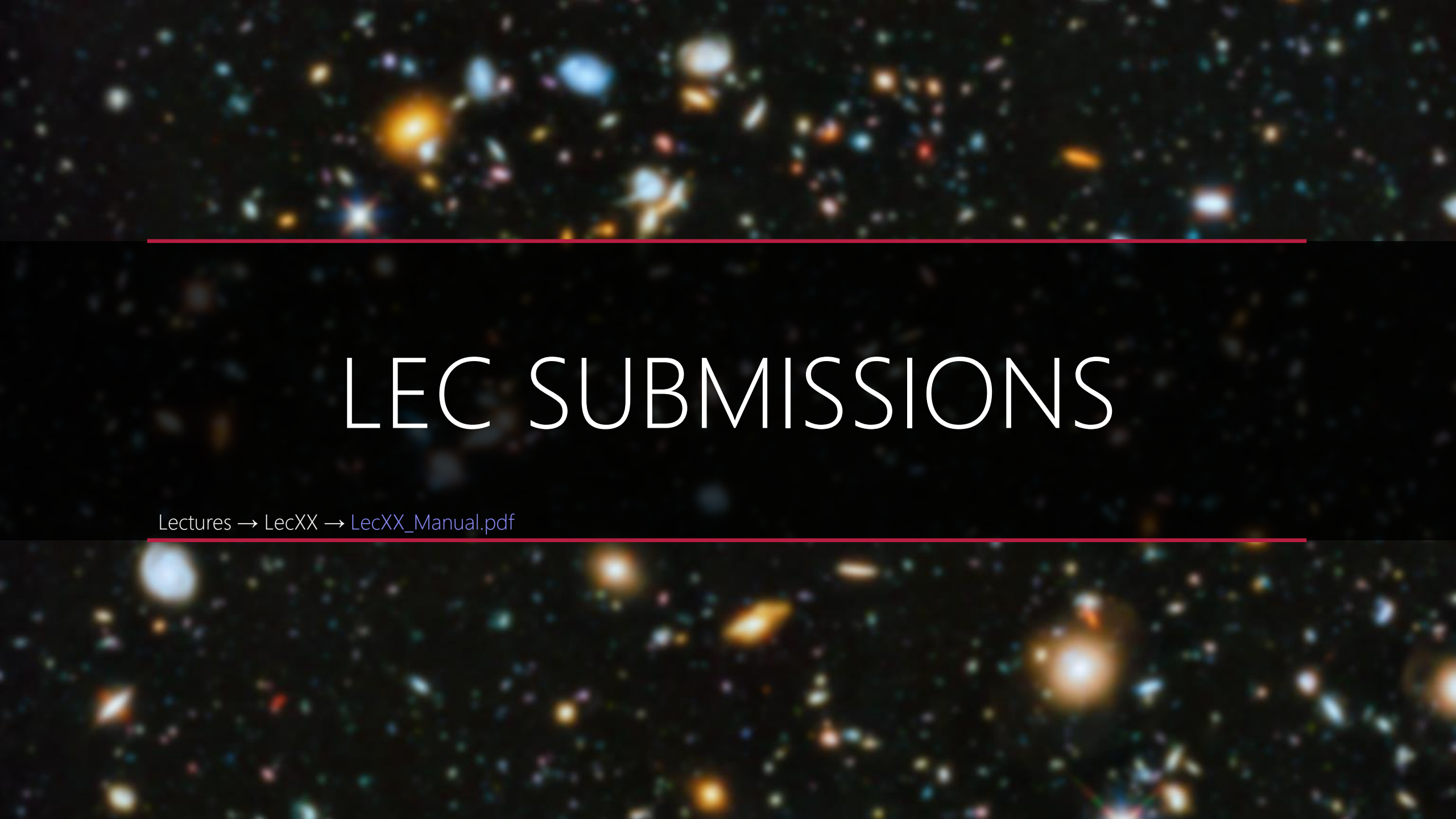
Jaber Al Siam (siamj@uwindsor.ca)

#51: Monday 5:30PM-6:50PM, Odette Building B06



Rustem Izmailov (izmailor@uwindsor.ca)

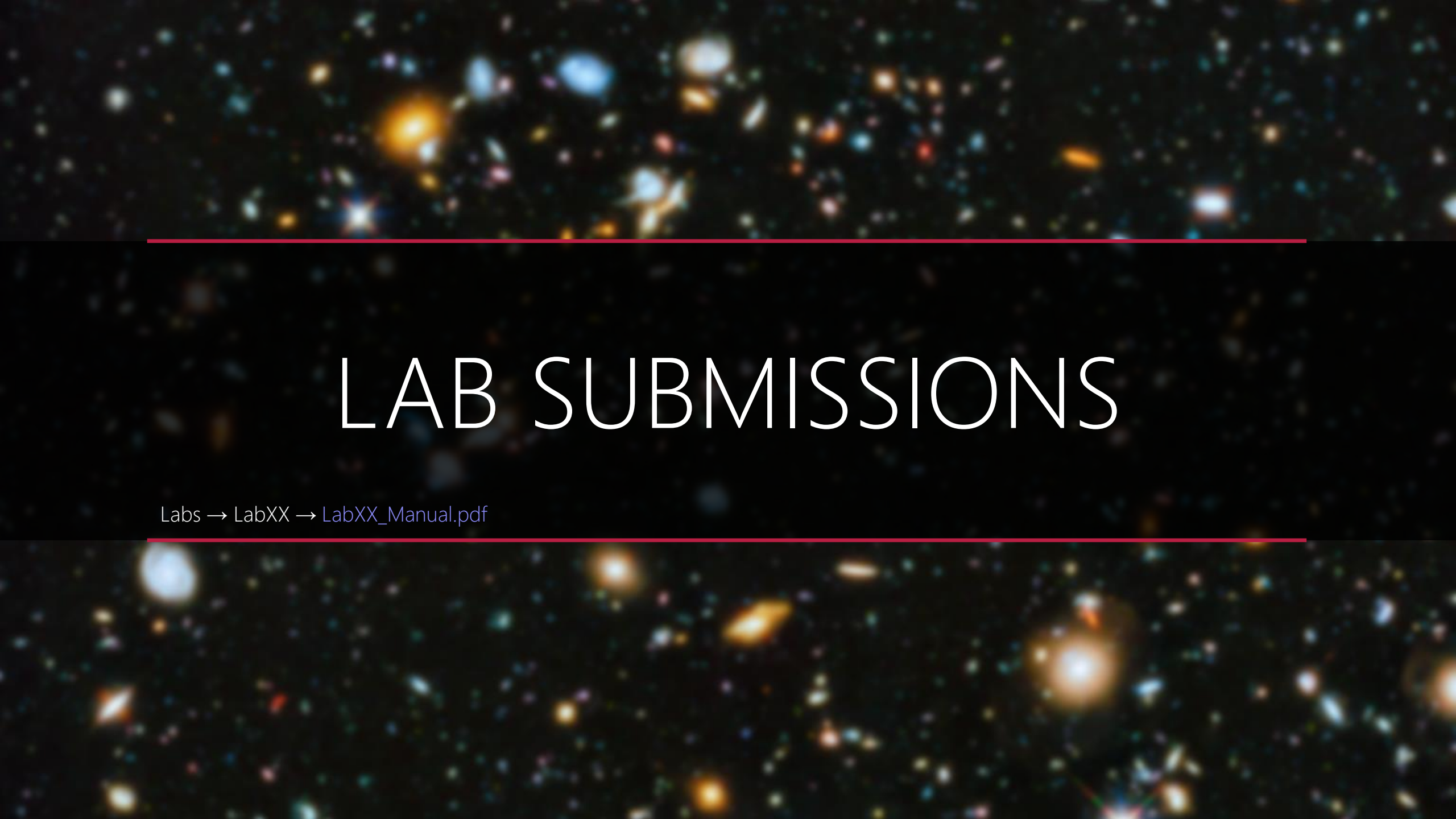
#52: Wednesday 5:30PM-6:50PM, Odette Building B06

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, fuzzy blobs and others as more distinct, elongated structures. A thin red horizontal line is positioned above the title.

LEC SUBMISSIONS

Lectures → LecXX → [LecXX_Manual.pdf](#)

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, fuzzy blobs and others as more distinct, elongated structures. A thin red horizontal line is positioned below the text.



LAB SUBMISSIONS

Labs → LabXX → [LabXX_Manual.pdf](#)

A cosmic background image featuring a dense field of galaxies and stars. The galaxies are in various stages of evolution, with some appearing as bright, irregular shapes and others as more structured, spiral or elliptical forms. The colors range from deep blues and purples to bright yellows and oranges, set against a dark, star-filled sky.

DISCUSSION BOARD

+ Bonus 5%

A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

SUPPLEMENTS

NOTETAKERS NEEDED

**VOLUNTEER
TO BE A
NOTETAKER**

We are looking
for a volunteer
notetaker for this
class. If you would
like to volunteer
please email
notetaker@uwindSOR.ca



master 3 Branches 2 Tags

hosseinfani	w12-w13. end of w2022	343663b · 3 years ago	64 Commits
Assistants	Winter 2022, up to W12	3 years ago	
Labs	Winter 2022, up to W12	3 years ago	
Lectures	w12-w13. end of w2022	3 years ago	
Misc	Add files via upload	3 years ago	
.gitattributes	Initial commit	5 years ago	
.gitignore	no message	4 years ago	
Book.jpg	Initial Commit	5 years ago	
Course_Syllabus.docx	Winter 2022, up to W12	3 years ago	
Course_Syllabus.pdf	Winter 2022, up to W12	3 years ago	
LICENSE.txt	License added	5 years ago	
Learning_Outcome.docx	W01 Winter 2021 Prep.	5 years ago	
Learning_Outcome.pdf	W01 Winter 2021 Prep.	5 years ago	
README.md	Winter 2022, up to W12	3 years ago	
Schedule_Timeline.png	Winter 2022, up to W12	3 years ago	
Schedule_Timeline.xlsx	Winter 2022, up to W12	3 years ago	

About

Materials for the Computer Science course, Digital Design (Logic Circuits)

- counter
- memory
- decoding
- karnaugh-map
- logic-gates
- boolean-algebra
- multiplexer
- adder
- shift-register
- registers
- digital-design
- digital-systems
- canonical-forms
- flip-flop
- combinational-logic
- sequential-logic
- number-systems
- subtractor
- logic-circuits
- programmable-logic

- Readme
- View license
- Activity
- 7 stars
- 1 watching
- 4 forks
- Report repository

Releases 2

Winter2021 at UWindsor Latest on Jan 10, 2022

+ 1 release

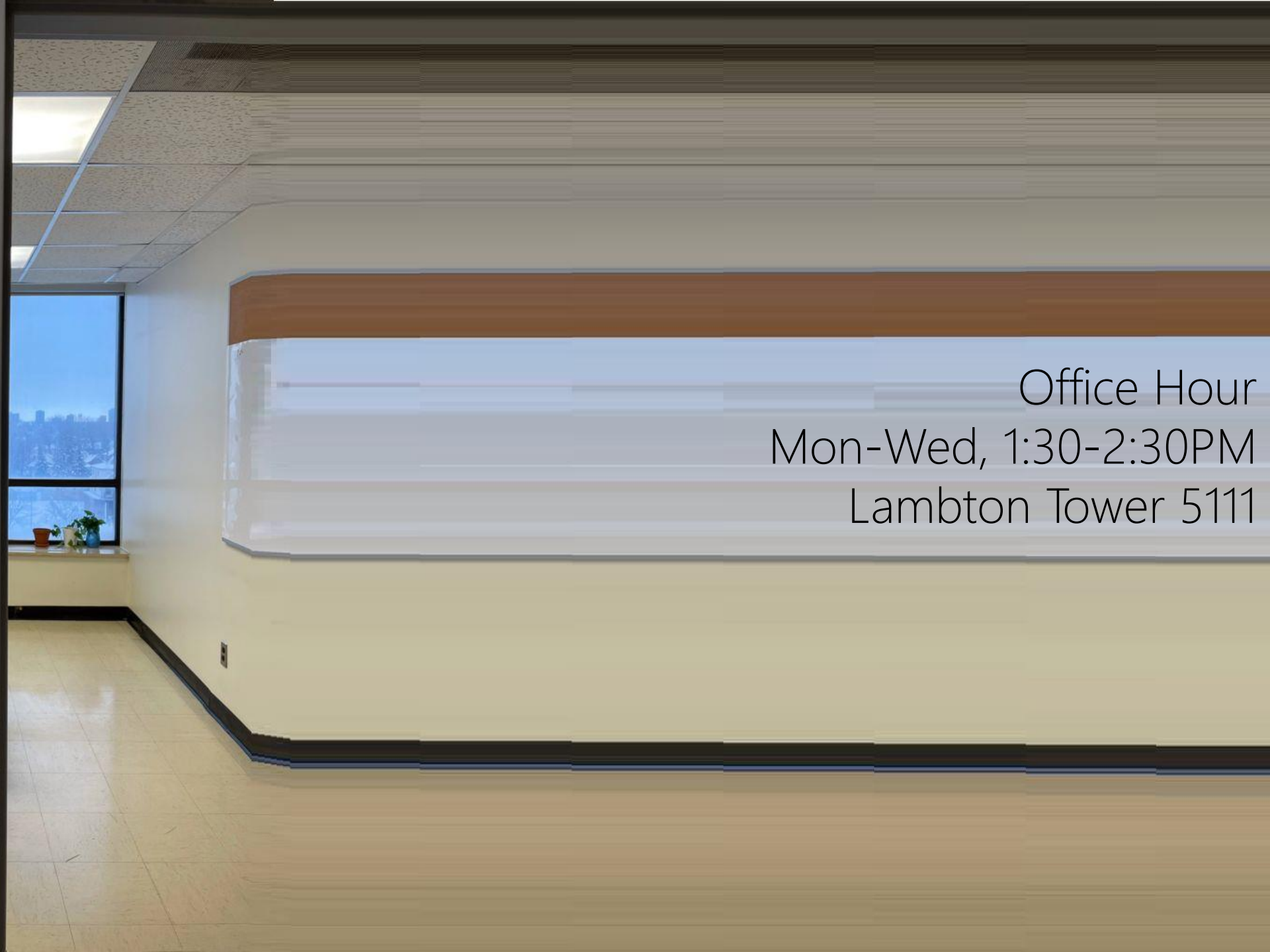


Hossein Fani, PhD

Assistant Professor, School of Computer Science, Faculty of Science, University of Windsor
Room 5111, Lambton Tower

hfani@uwindsor.ca

cs.uwindsor.ca/~hfani



Office Hour
Mon-Wed, 1:30-2:30PM
Lambton Tower 5111

- Change of classroom and labrooms
- Assignments
 - For marking and any related issue, foremost reach out to your lab instructor, not the marker. Then me!
 - For opening another submissions
 - Pay attention to naming of files!
 - Some topics are not covered in the lecture, but in labs or lecture assignments
 - Lab may have dependencies.
 - No key to the labs!
 - Key to the assignments based on highest selected assignments
 - Questions about the lab or lec in discussion board.
 - Subscribe to the forums to receive emails when somebody put sth
 - Academic Integrity (plagiarism)
- Keys to the exams (midterm and final)

- Blackboard is unavailable every weekdays (Monday to Friday) from 5 AM to 6 AM for regular maintenance!
- Change of classroom and labrooms
- Assignments
 - For marking and any related issue, foremost reach out to your lab instructor, not the marker. Then me!
 - For opening another submissions
 - Pay attention to version of files

SLIDES ARE NOT LIKE THIS

- No key to the labs.
 - Key to the assignments based on highest selected assignments
 - Questions about the lab or lec in discussion board.
 - Subscribe to the forums to receive emails when somebody put sth
- Keys to the exams (midterm and final)

Slides are not books, summaries, ...
Slides are not for last night before exam
Slides are to be presented! Lecture = Slide + Talk

Are robots becoming more human or humans becoming more robotic?
<https://www.youtube.com/watch?v=YR5ApYxkU-U>



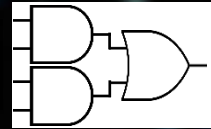
The Bonnie Situation, Pulp Fiction (1994), Quentin Tarantino

Number Systems $|(12)_{10} \rightarrow (1100)_2$

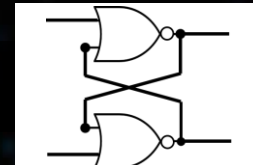
Logic Gates |



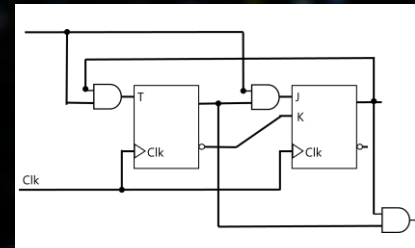
Combinational Logic |

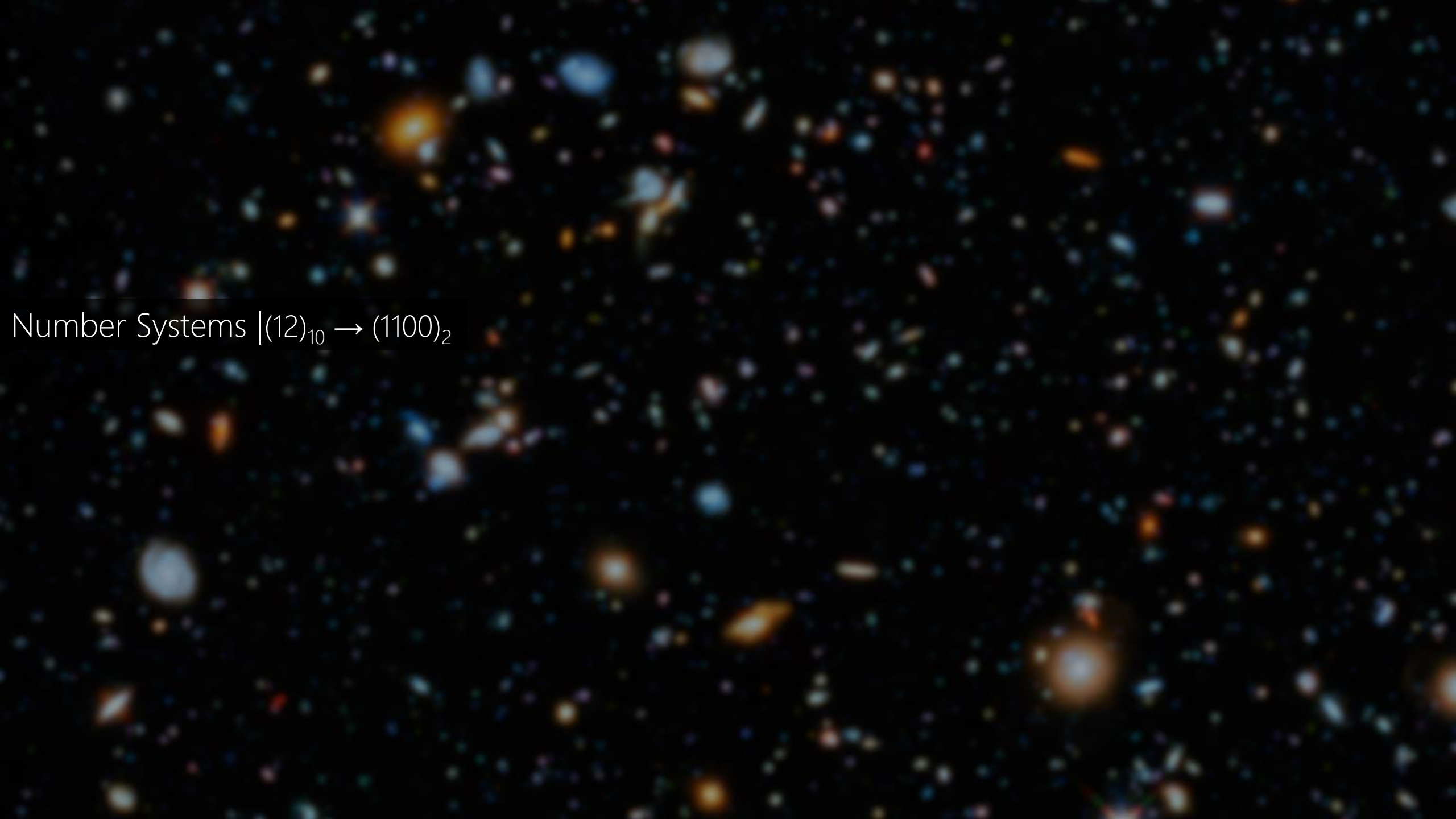


Flip-Flop |

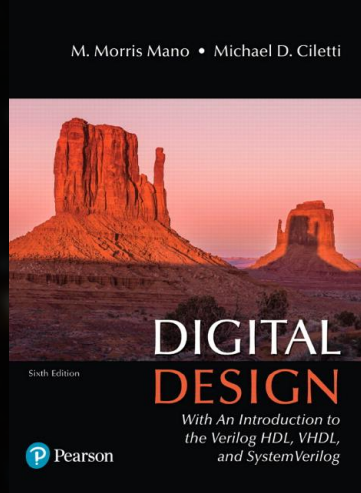


Sequential Logic |
Computers



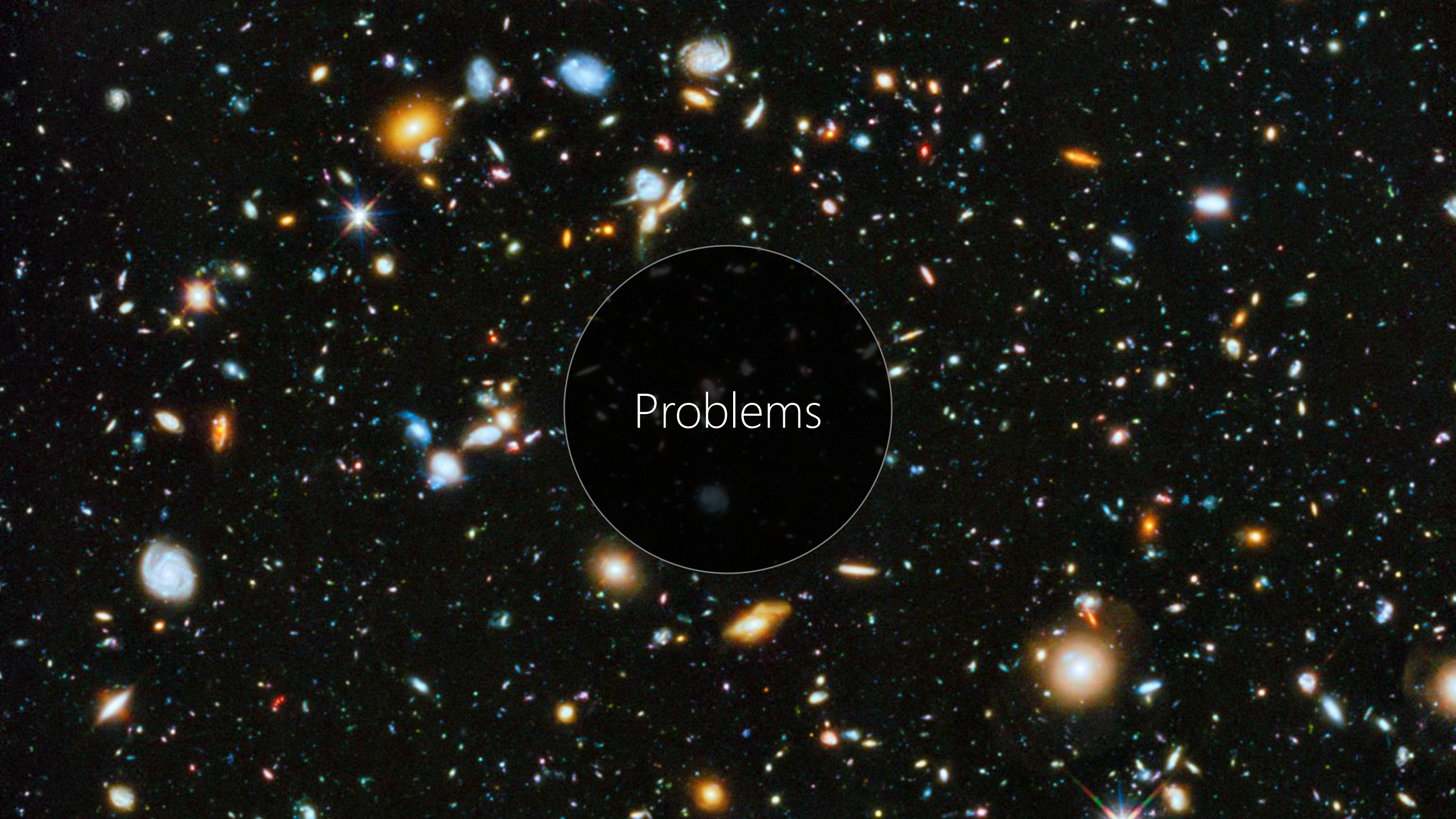


Number Systems $|(12)_{10} \rightarrow (1100)_2$

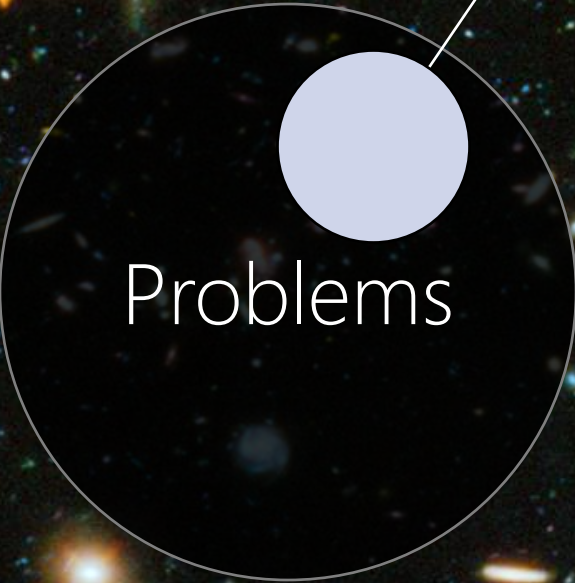
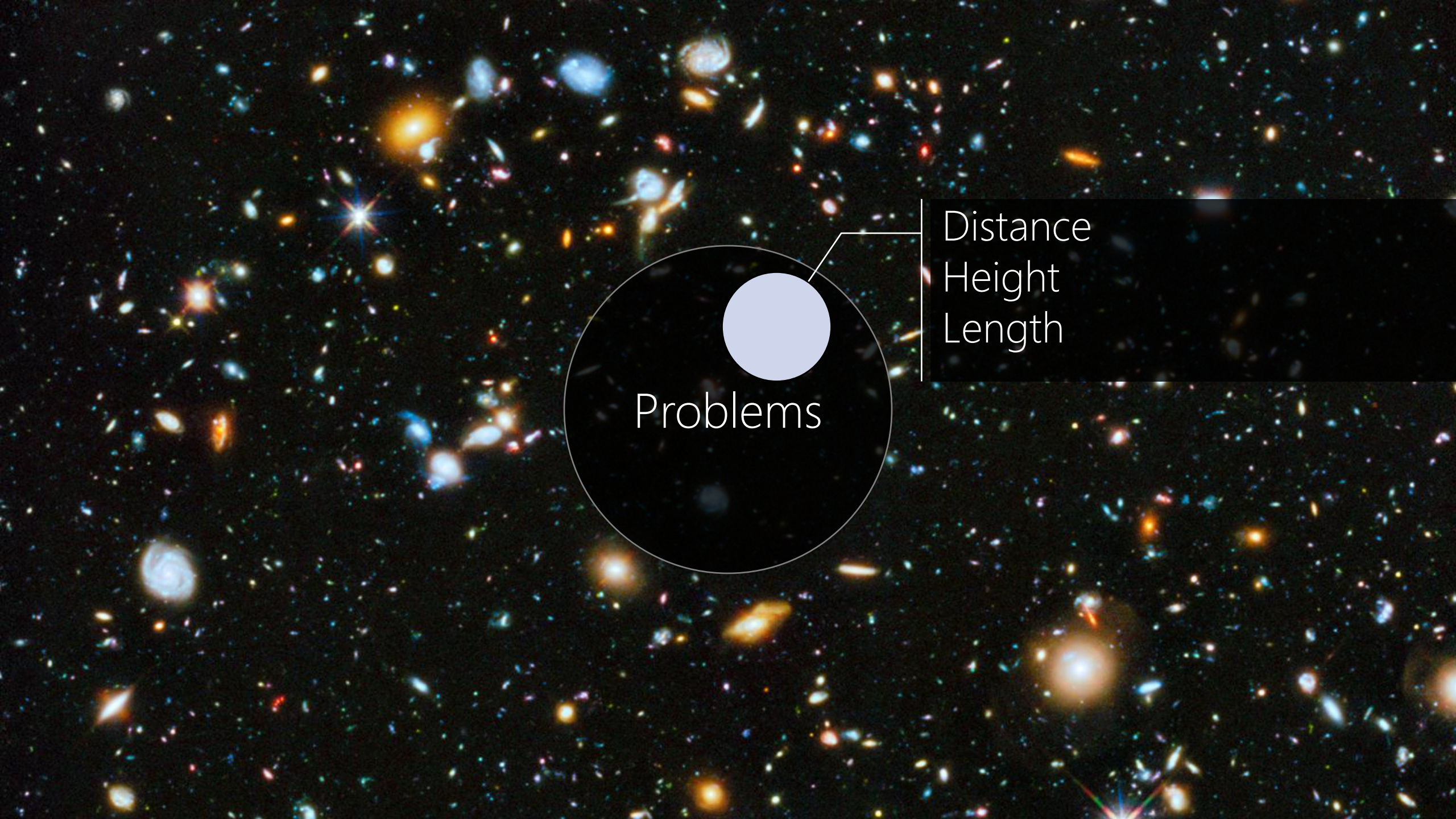


Chapter 1

Digital Systems and Binary Numbers

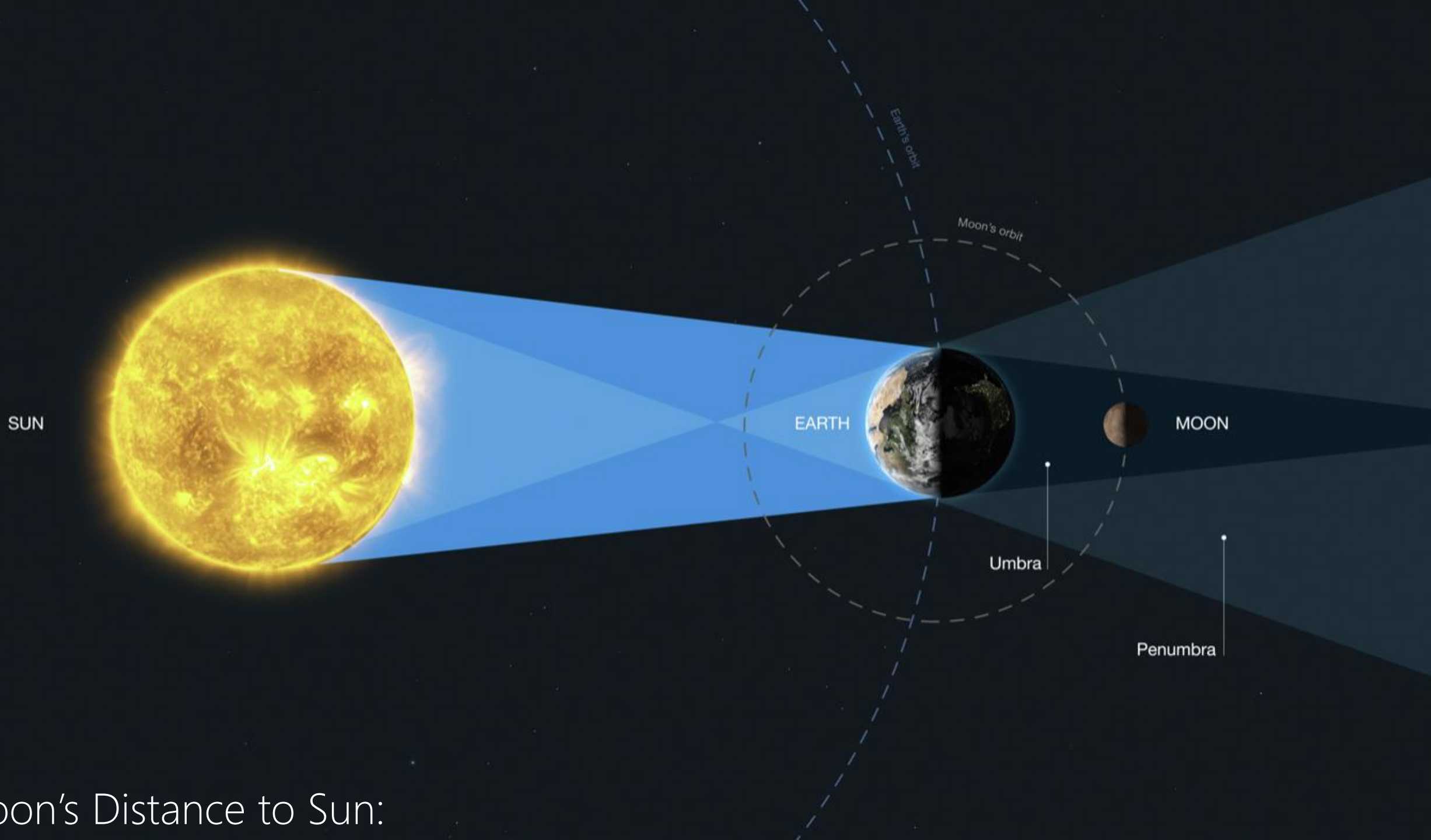
A deep-field astronomical image showing a vast number of galaxies in various colors (blue, orange, white) against a black background. The galaxies are of different shapes and sizes, some appearing as bright, distinct objects while others are fainter. A large, solid black circle is centered in the image, containing the word "Problems" in white text.

Problems

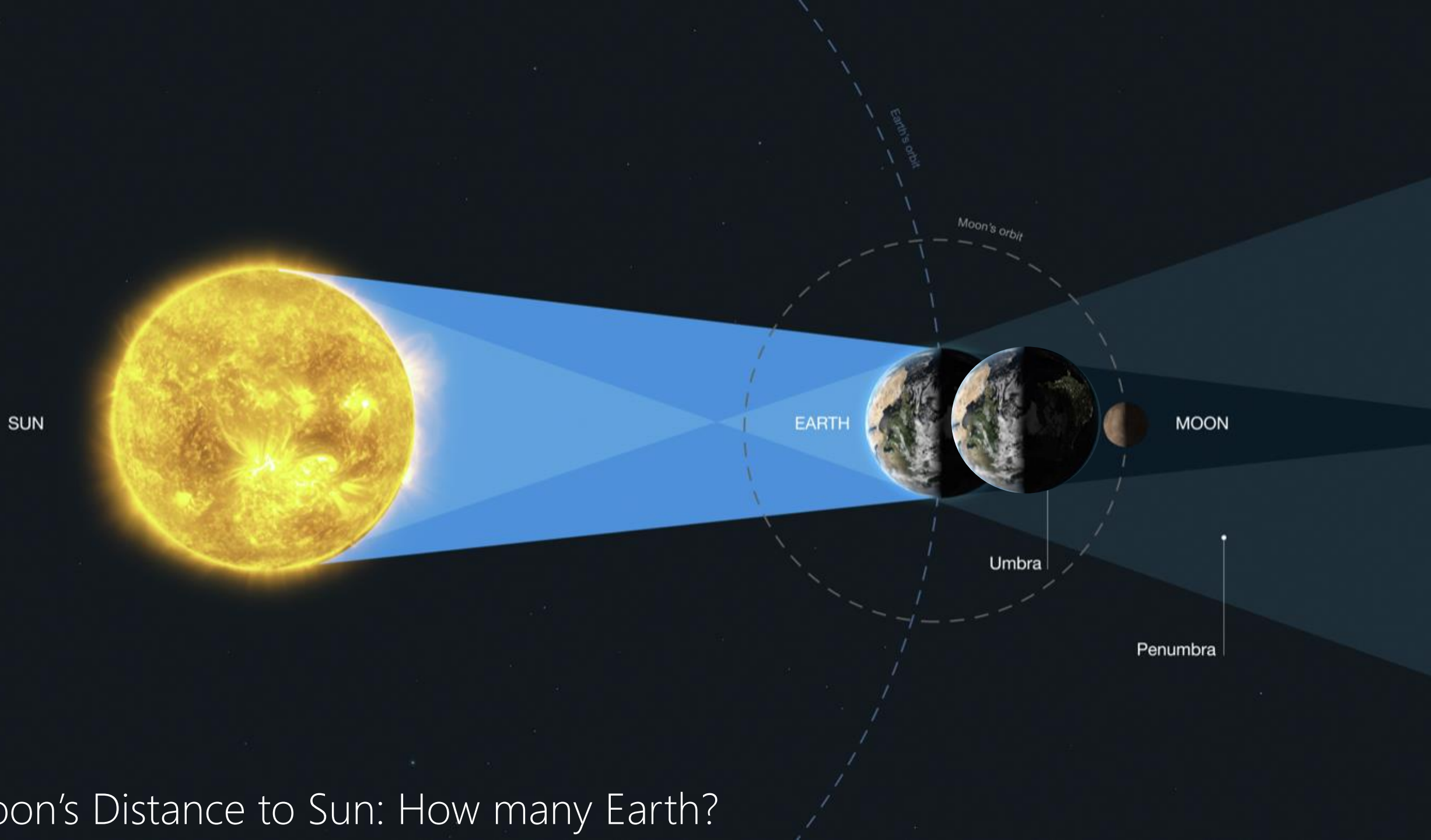


Distance
Height
Length

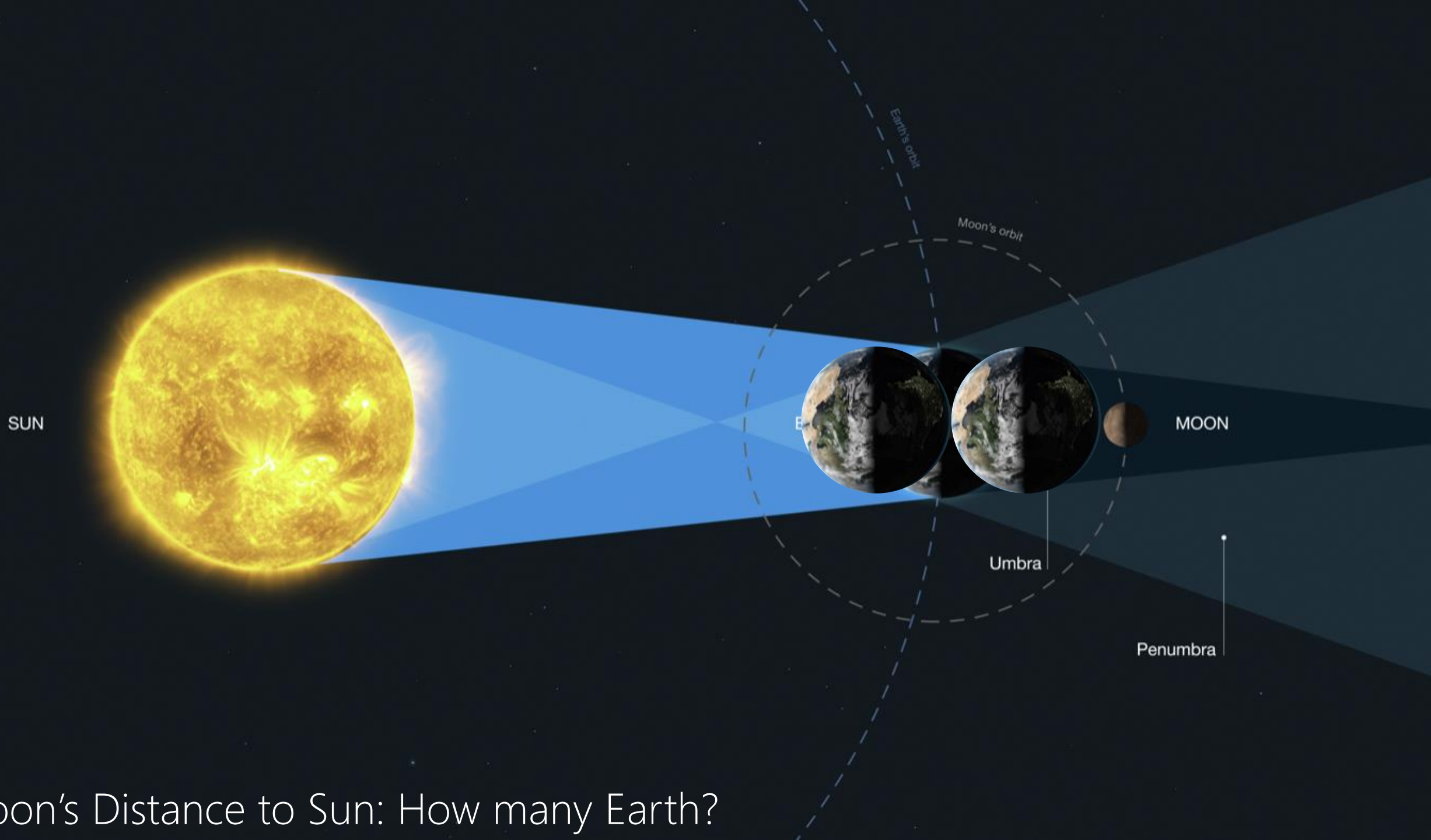
Problems



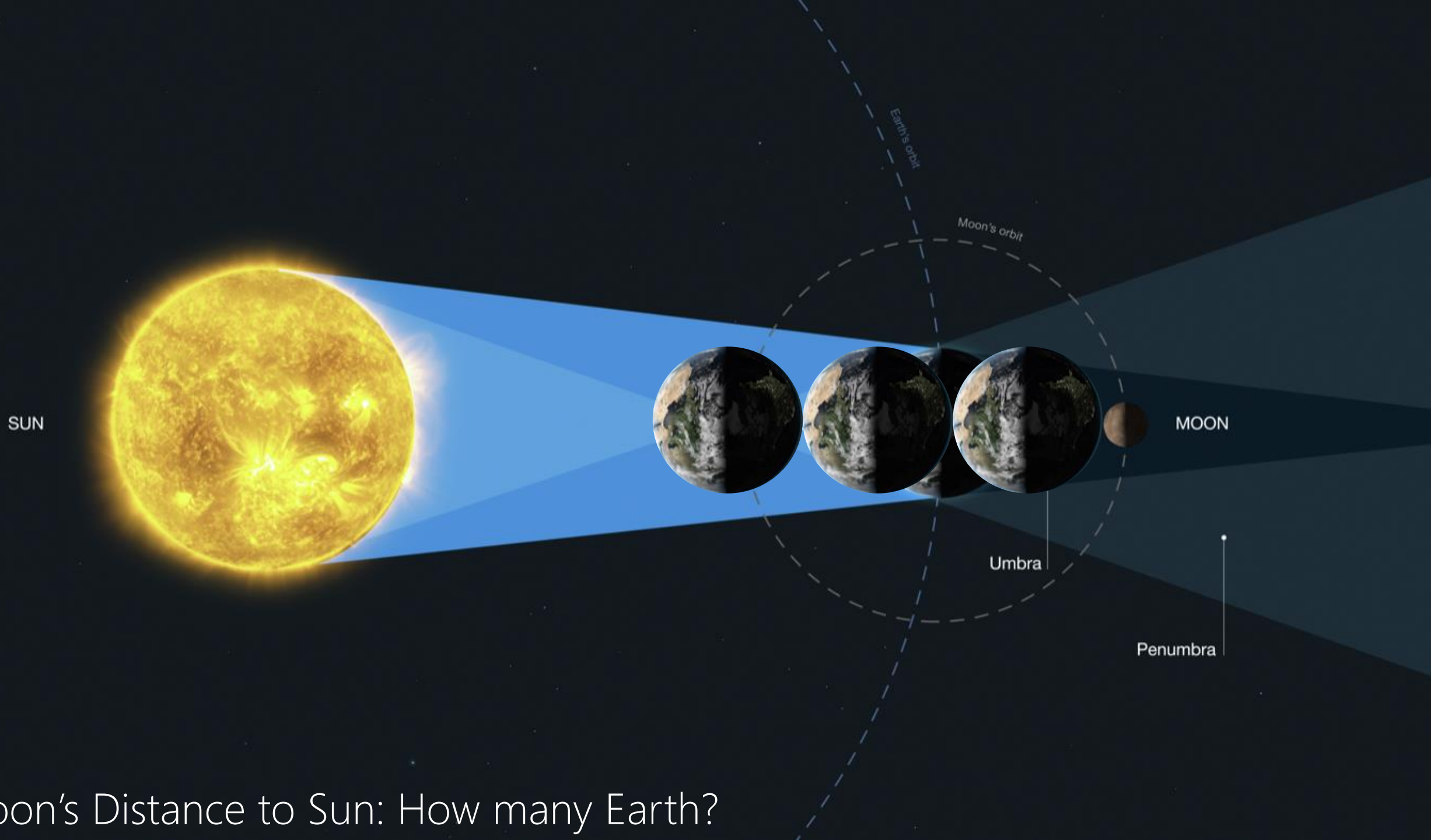
Moon's Distance to Sun:



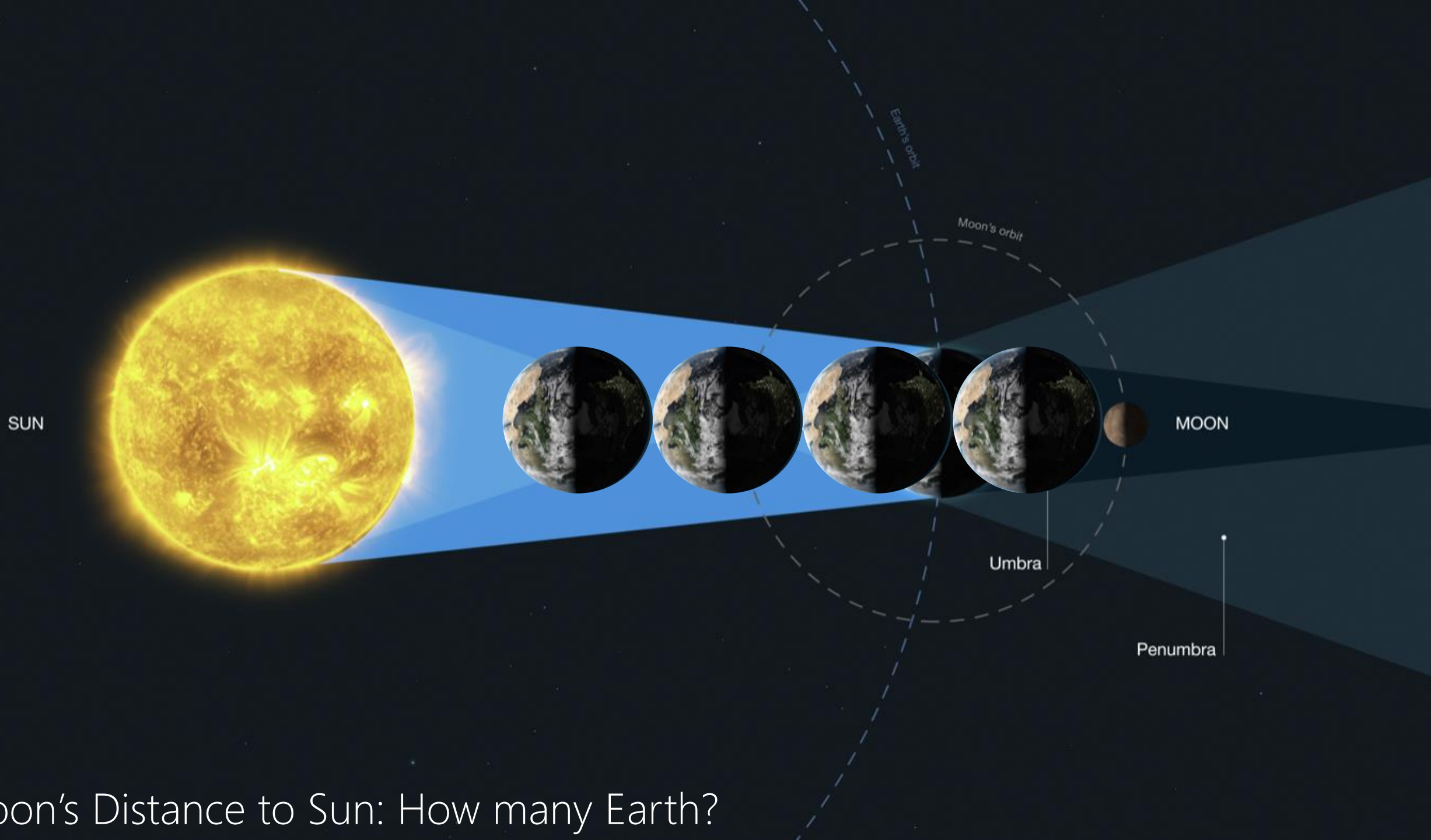
Moon's Distance to Sun: How many Earth?



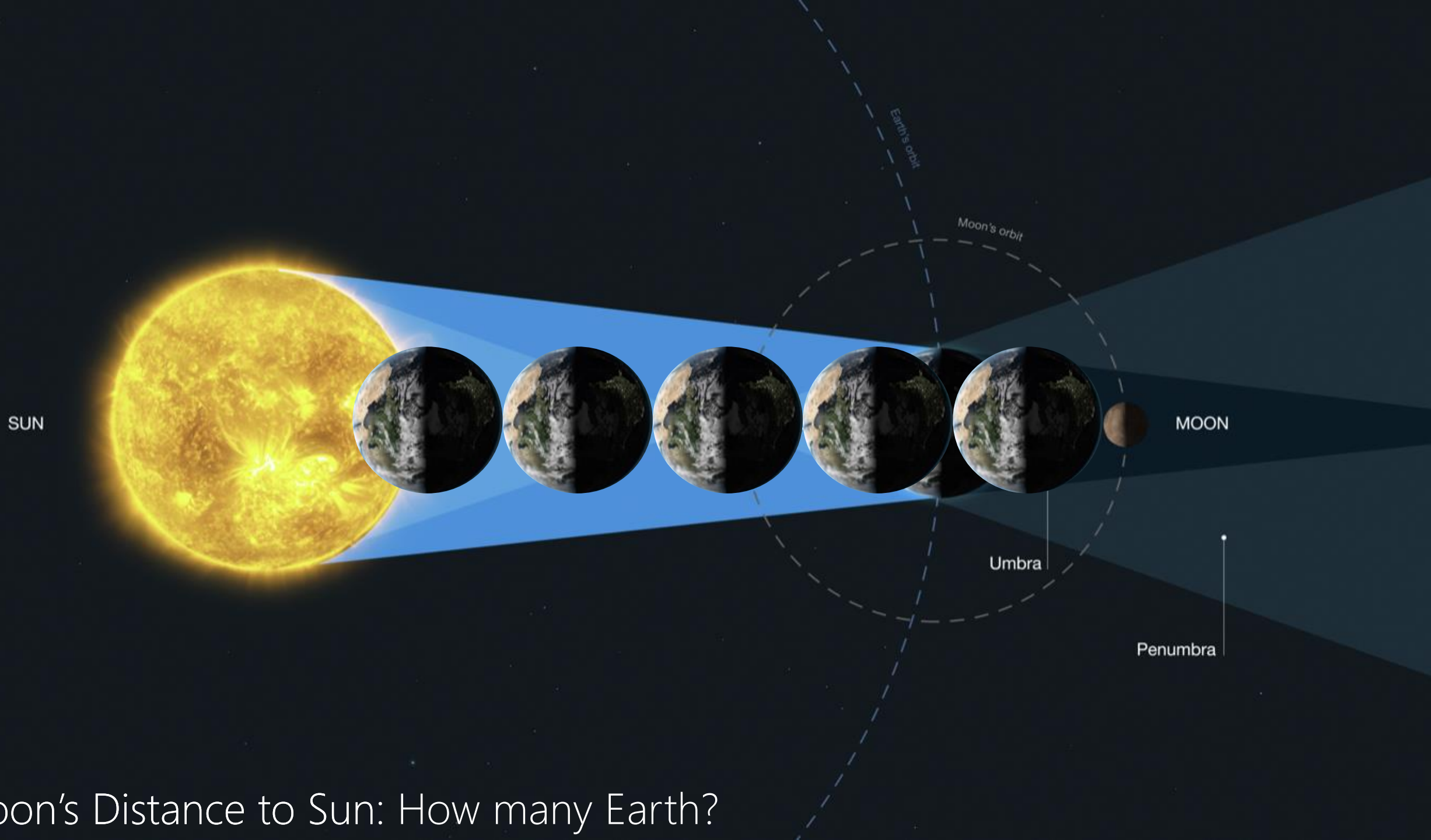
Moon's Distance to Sun: How many Earth?



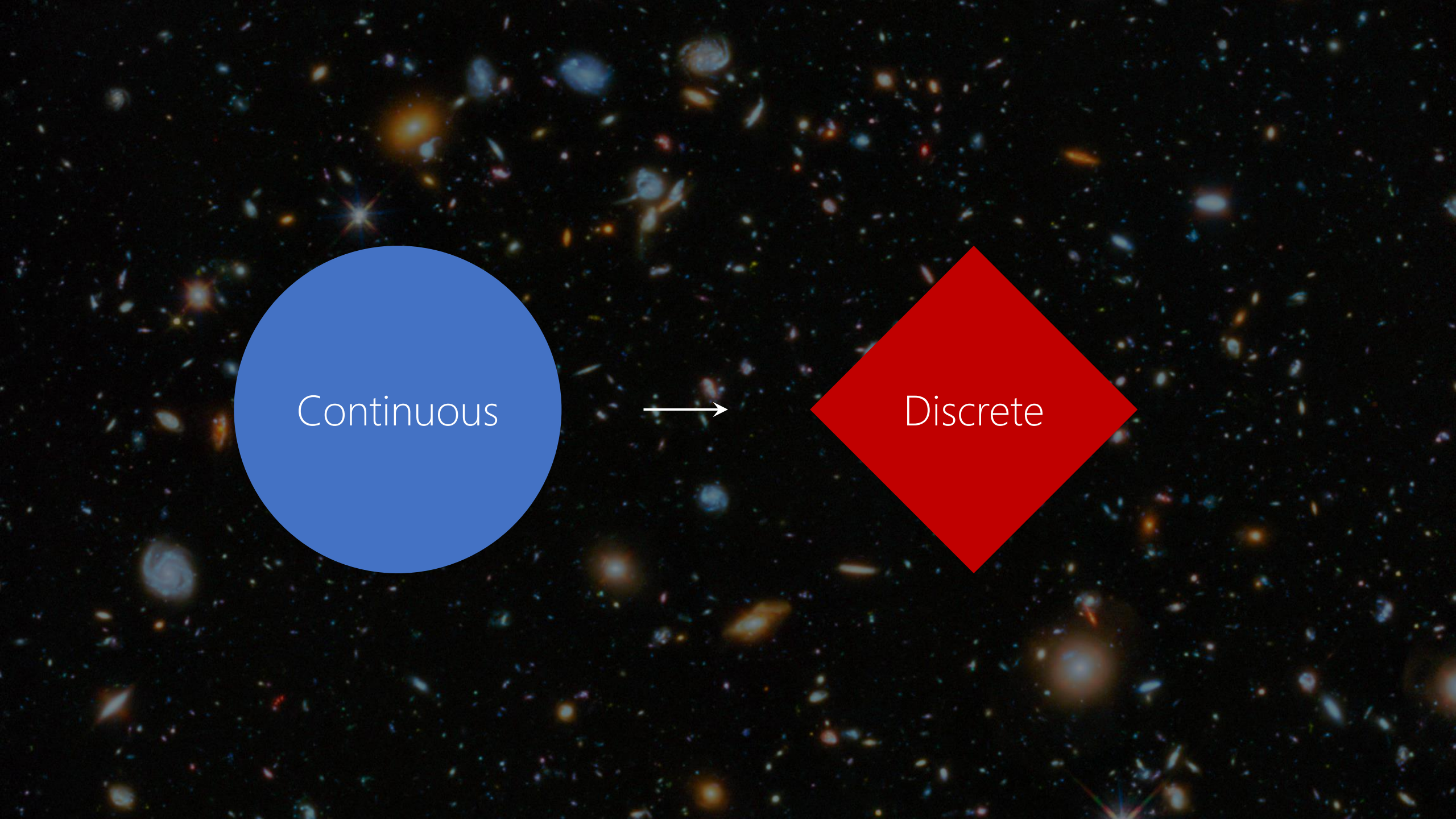
Moon's Distance to Sun: How many Earth?



Moon's Distance to Sun: How many Earth?



Moon's Distance to Sun: How many Earth?



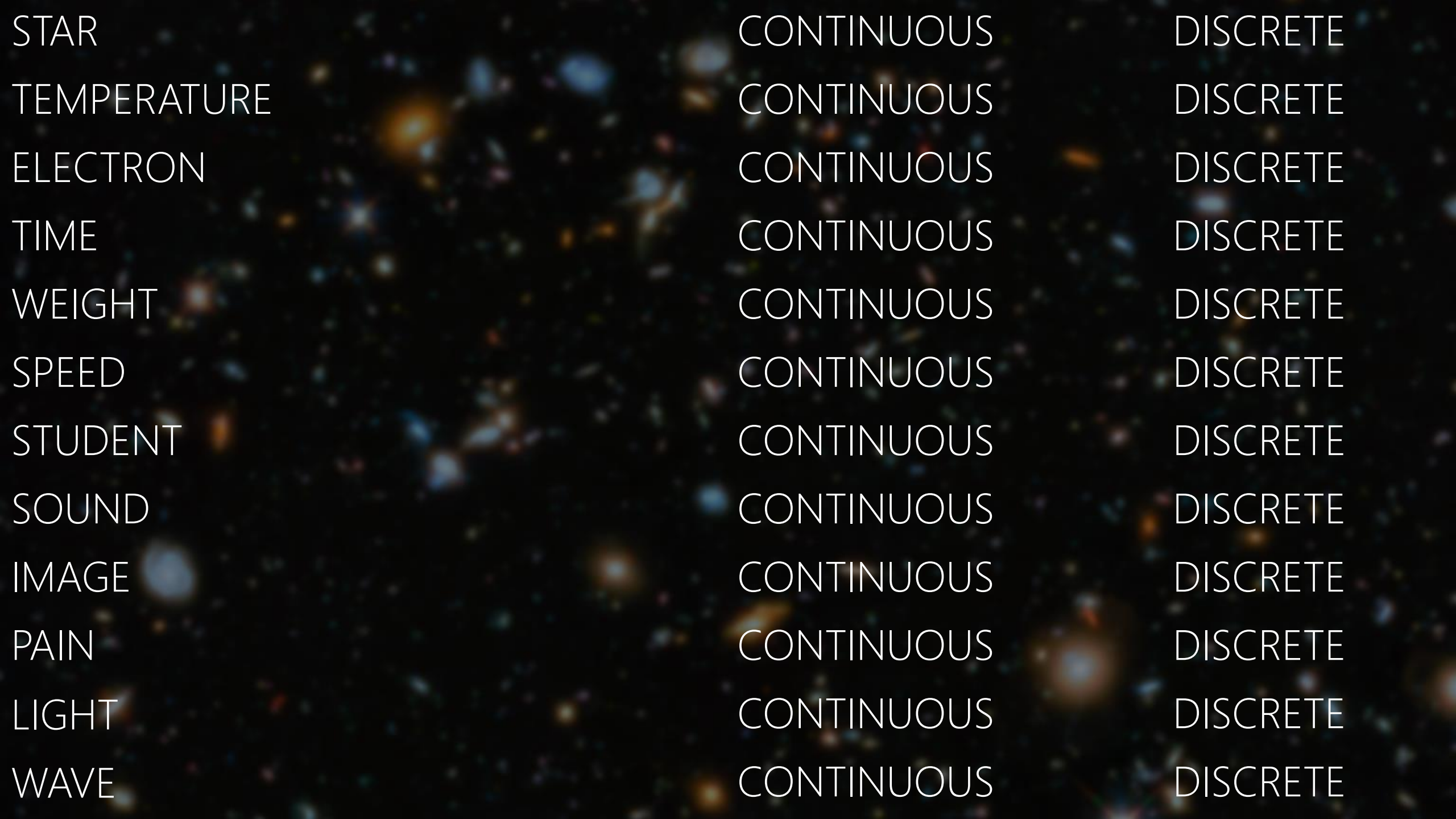
Continuous



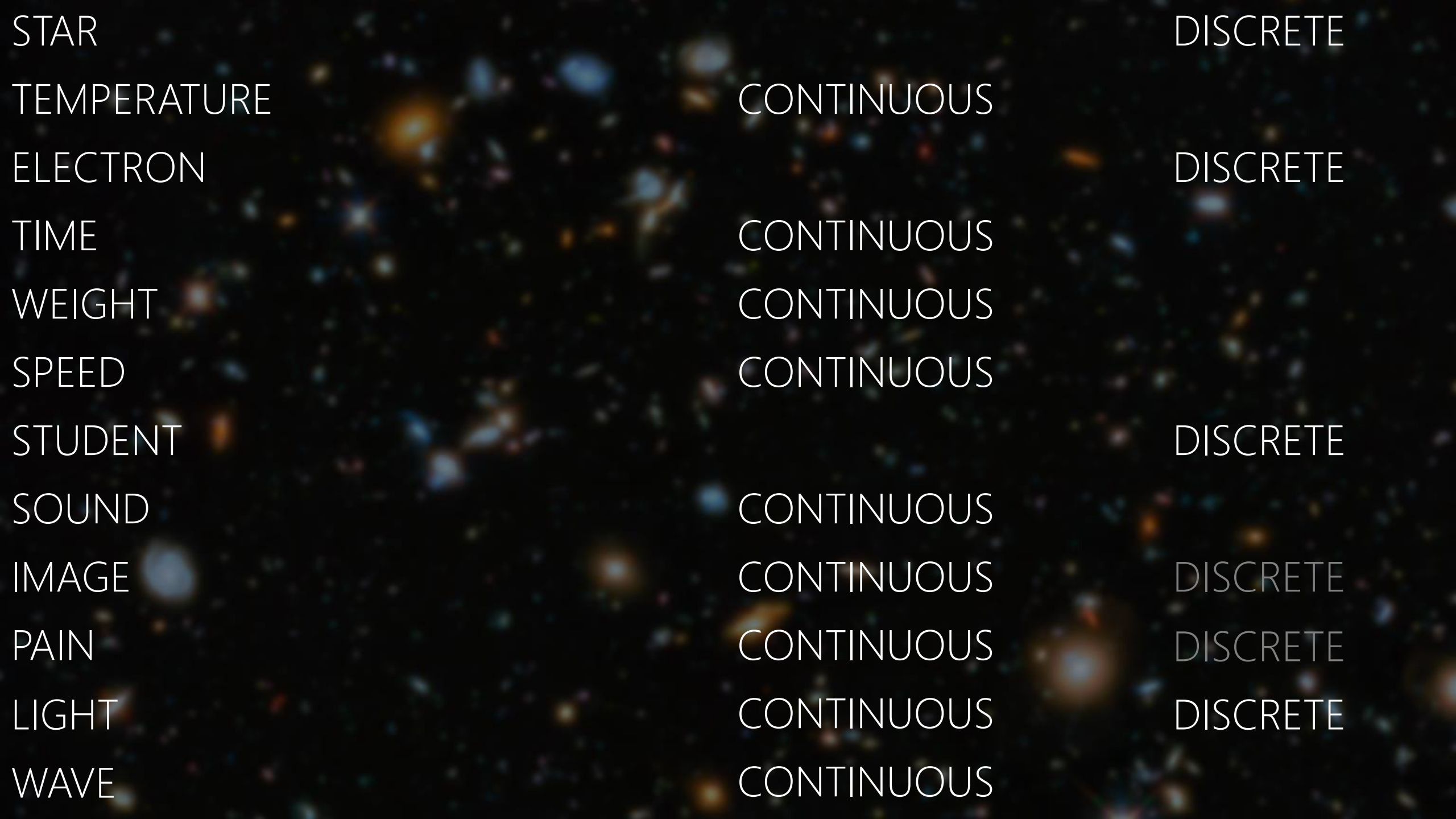
Discrete



DISCRETE SYSTEMS



STAR	CONTINUOUS	DISCRETE
TEMPERATURE	CONTINUOUS	DISCRETE
ELECTRON	CONTINUOUS	DISCRETE
TIME	CONTINUOUS	DISCRETE
WEIGHT	CONTINUOUS	DISCRETE
SPEED	CONTINUOUS	DISCRETE
STUDENT	CONTINUOUS	DISCRETE
SOUND	CONTINUOUS	DISCRETE
IMAGE	CONTINUOUS	DISCRETE
PAIN	CONTINUOUS	DISCRETE
LIGHT	CONTINUOUS	DISCRETE
WAVE	CONTINUOUS	DISCRETE



STAR		DISCRETE
TEMPERATURE	CONTINUOUS	
ELECTRON		DISCRETE
TIME	CONTINUOUS	
WEIGHT	CONTINUOUS	
SPEED	CONTINUOUS	
STUDENT		DISCRETE
SOUND	CONTINUOUS	
IMAGE	CONTINUOUS	DISCRETE
PAIN	CONTINUOUS	DISCRETE
LIGHT	CONTINUOUS	DISCRETE
WAVE	CONTINUOUS	

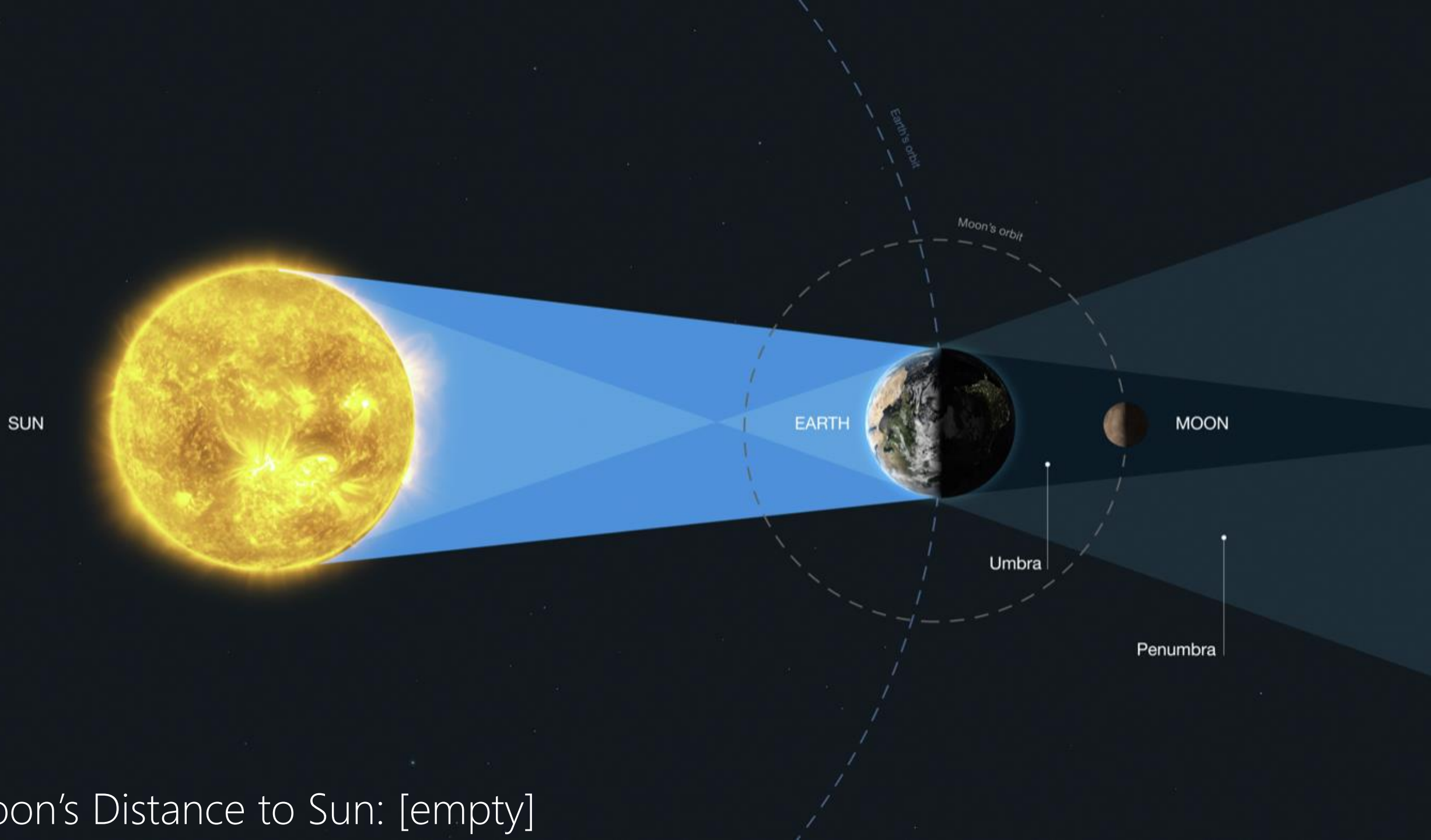


"The world's first photograph—or at least the oldest surviving photo—was taken by Joseph Nicéphore Niépce in 1826 or 1827. Captured using a technique known as heliography, the shot was taken from an upstairs window at Niépce's estate in Burgundy. As heliography produces one-of-a-kind images, there are no duplicates of the piece, which is now part of the permanent collection at the University of Texas-Austin." [18 Famous First Photographs in History: From the Oldest Photo Ever to the World's First Instagram](#)

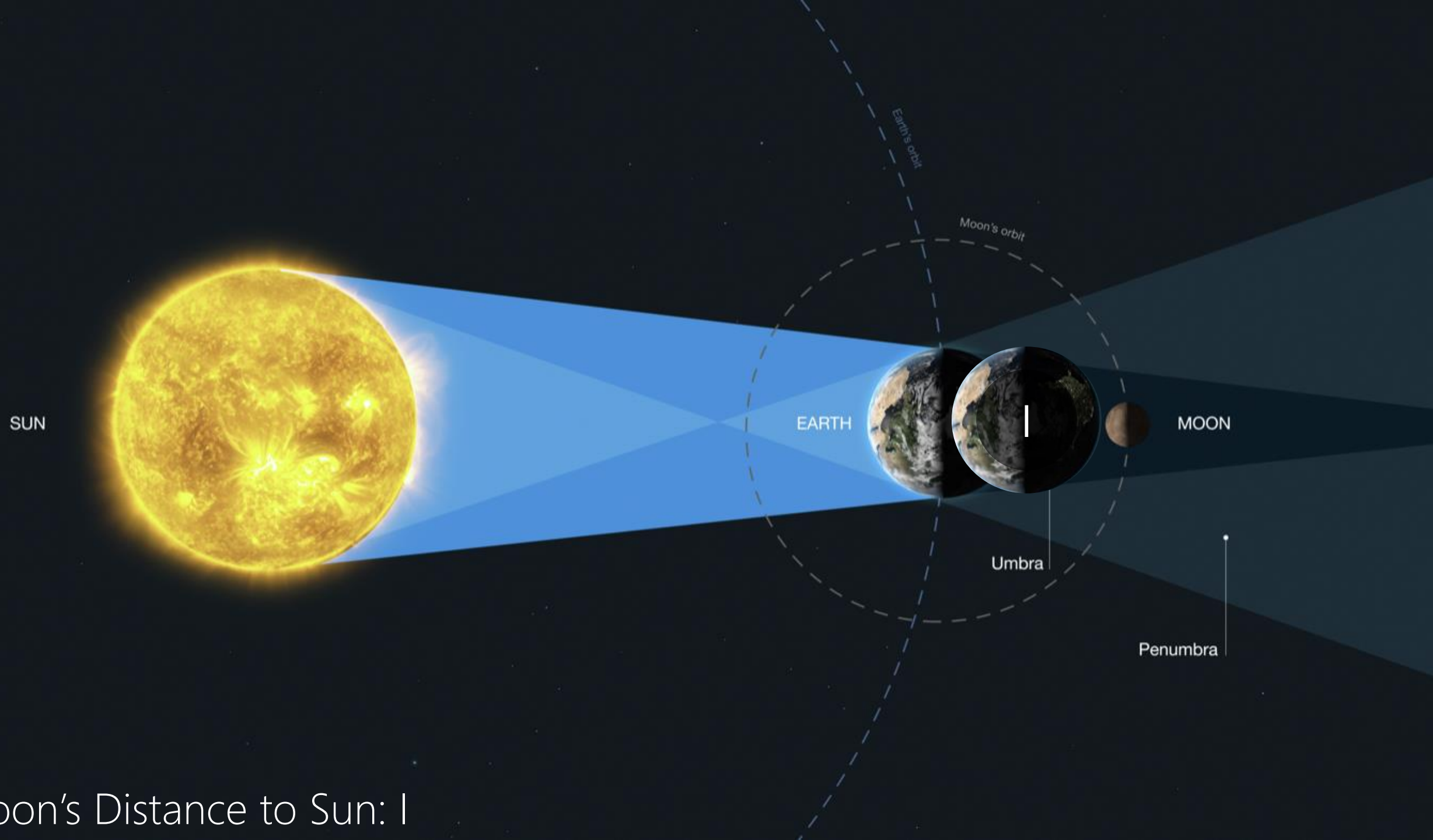




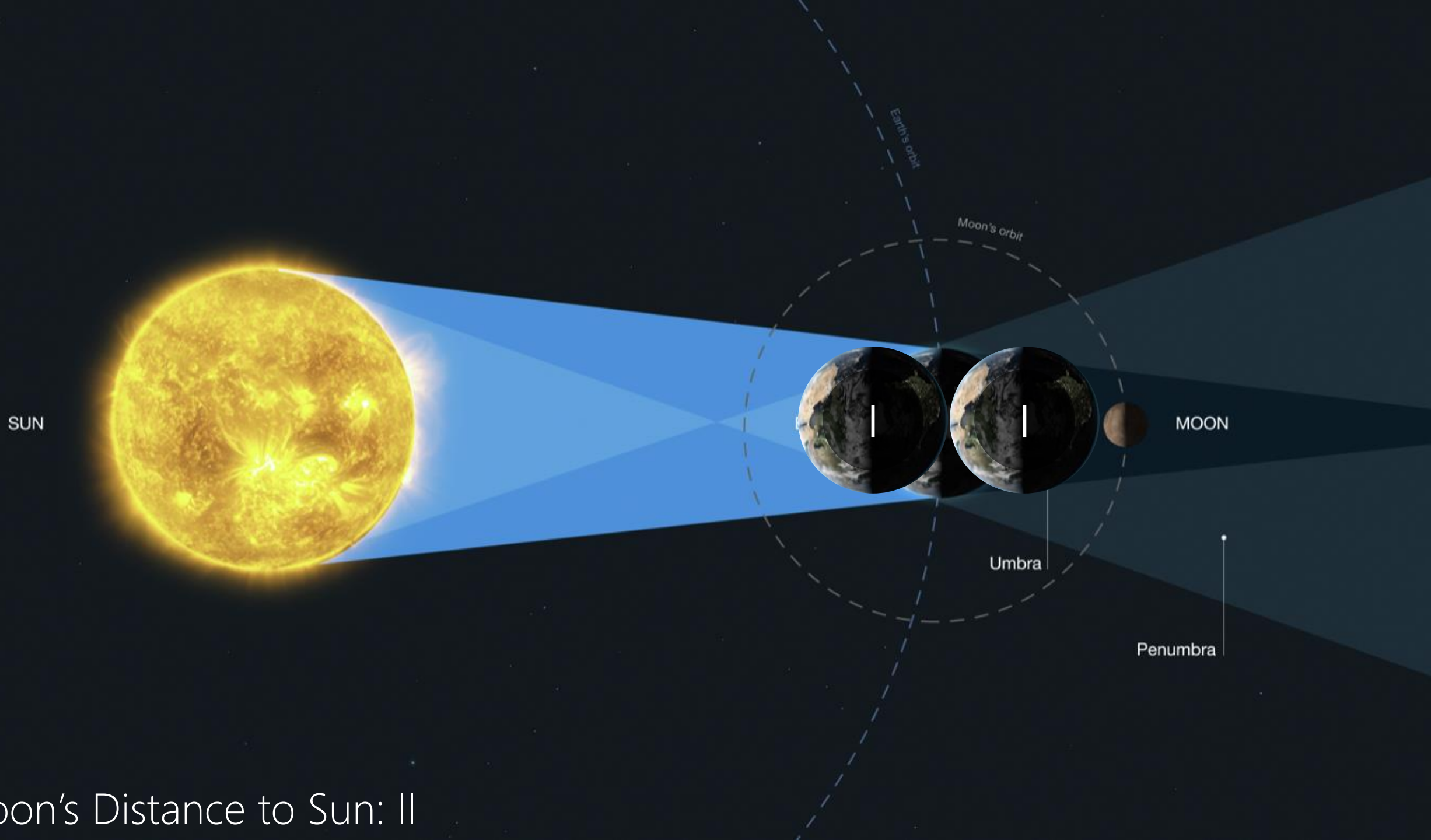
"We may be used to selfies now, but it's Robert Cornelius's 1839 image that lays claim to the first self-portrait. Taken in Philadelphia, Cornelius sat for a little over one minute before covering the lens." [18 Famous First Photographs in History: From the Oldest Photo Ever to the World's First Instagram](#)



Moon's Distance to Sun: [empty]

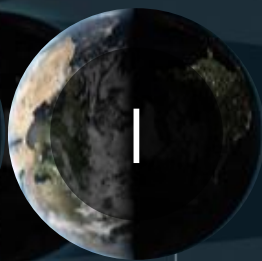
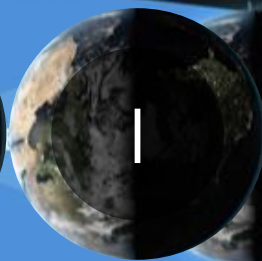
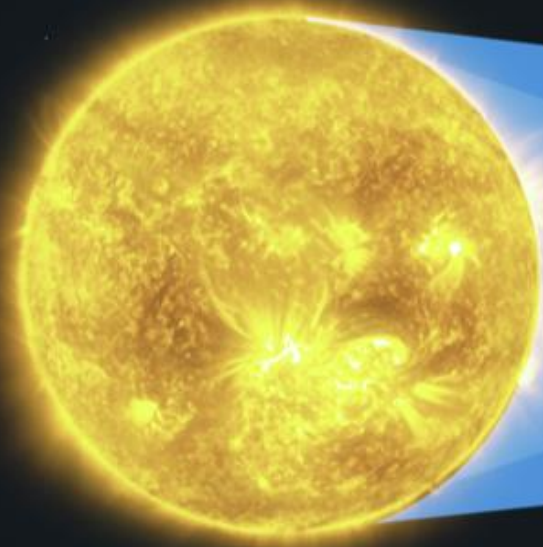


Moon's Distance to Sun: 1



Moon's Distance to Sun: II

SUN



MOON

Earth's orbit

Moon's orbit

Umbra

Penumbra

Moon's Distance to Sun: III

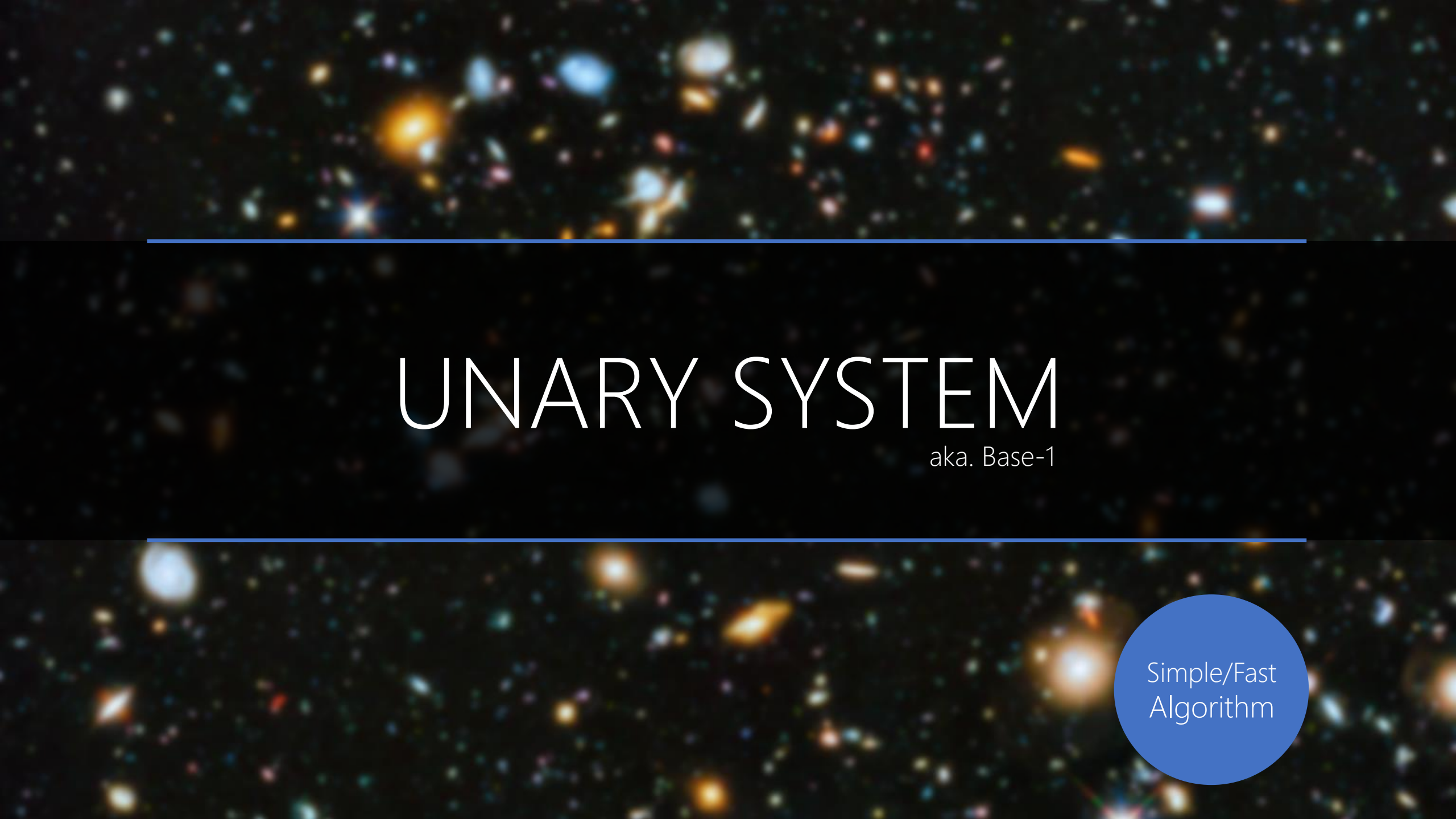


Quantization



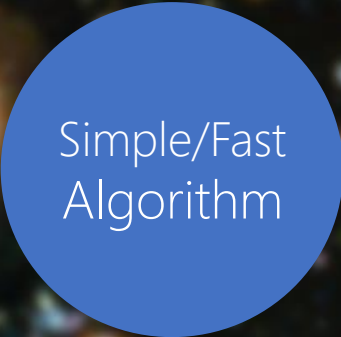
I II III IV V VI VII VIII IX X

Roman Numerals
Originated in Ancient Rome
8th Century BC

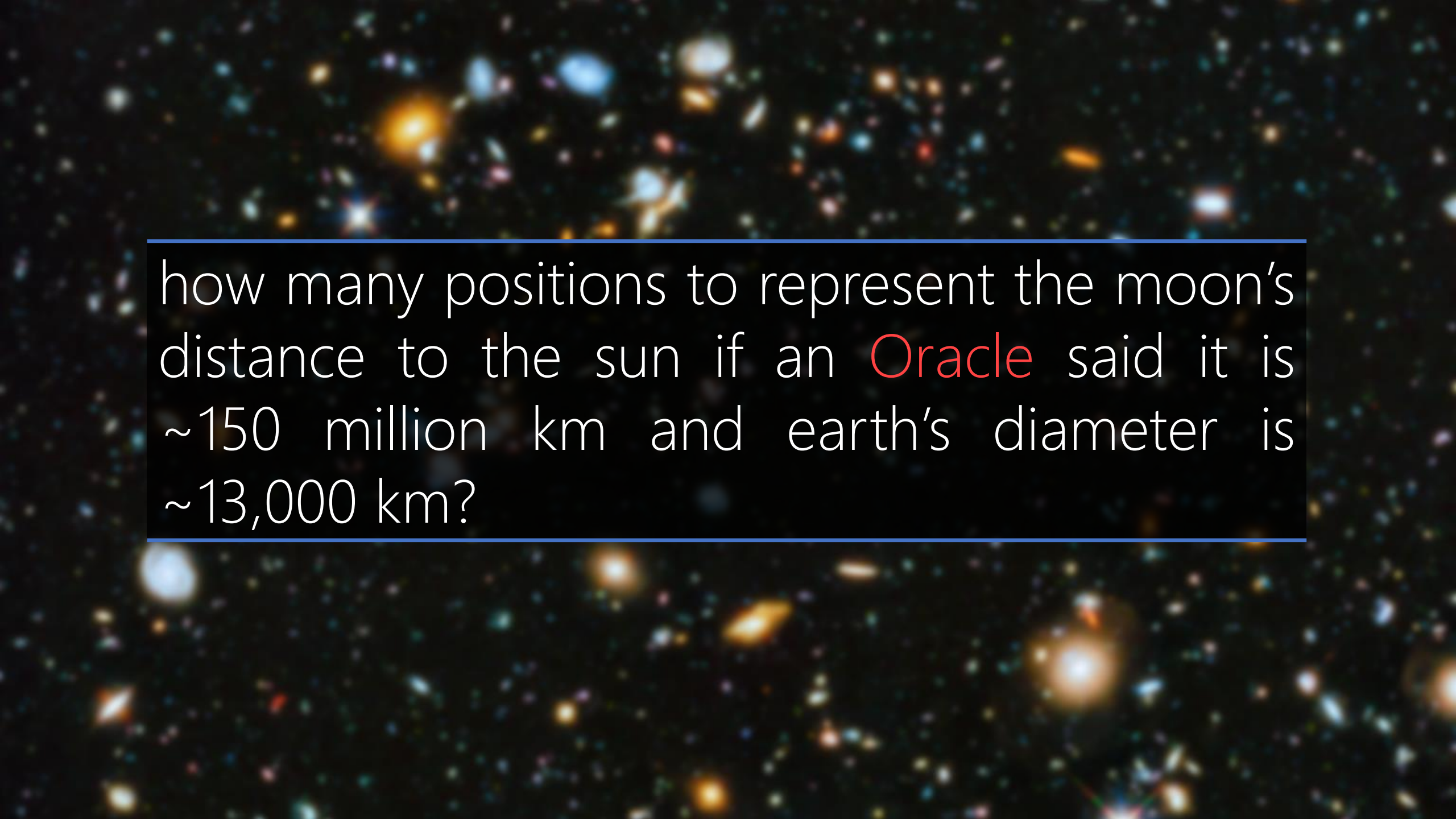


UNARY SYSTEM

aka. Base-1



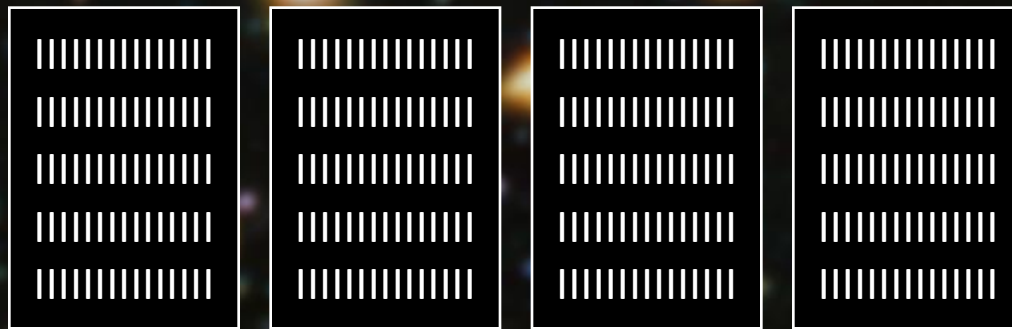
Simple/Fast
Algorithm



how many positions to represent the moon's distance to the sun if an **Oracle** said it is ~150 million km and earth's diameter is ~13,000 km?

$\sim 150 \text{ million km} \div \sim 13,000 \text{ km} = \sim 12,000 \text{ Earth}$
paper = $\sim 3,000 \text{ positions}$
 $12,000 \div 3,000 = 4 \text{ pages!}$

Poor Data
Model



A deep-field astronomical image showing a vast field of galaxies in various colors (blue, orange, white) against a black background. Two horizontal blue lines frame the central text.

NUMBER SYSTEMS

— = ≡ ≠ ୮ | ୬ ୭ ୫ ୨

Brahmi

3rd and 7th century AD

୨ ୩ ୪ ୫ ୬ | ୭ ୮ ୯ ୦

Hindu (Gwalior)

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯ ୦

Sanskrit-Devanagari

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯

Western Arabic (Gobar)

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯ ୦

Eastern Arabic

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯ ୦

11th Century (Apices)

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯ ୦

15th Century

୧ ୨ ୩ ୪ ୫ | ୬ ୭ ୮ ୯ ୦

16th Century (Dürer)



Play (k)

▶ ⏮ 🔊 2:31 / 13:29 • Introduction >

⏸ CC ⚙ 📺 🖥 🗑

Neil deGrasse Tyson Explains Zero



StarTalk ✓
3.67M subscribers

Subscribe

👍 8.5K



➦ Share

⬇ Download

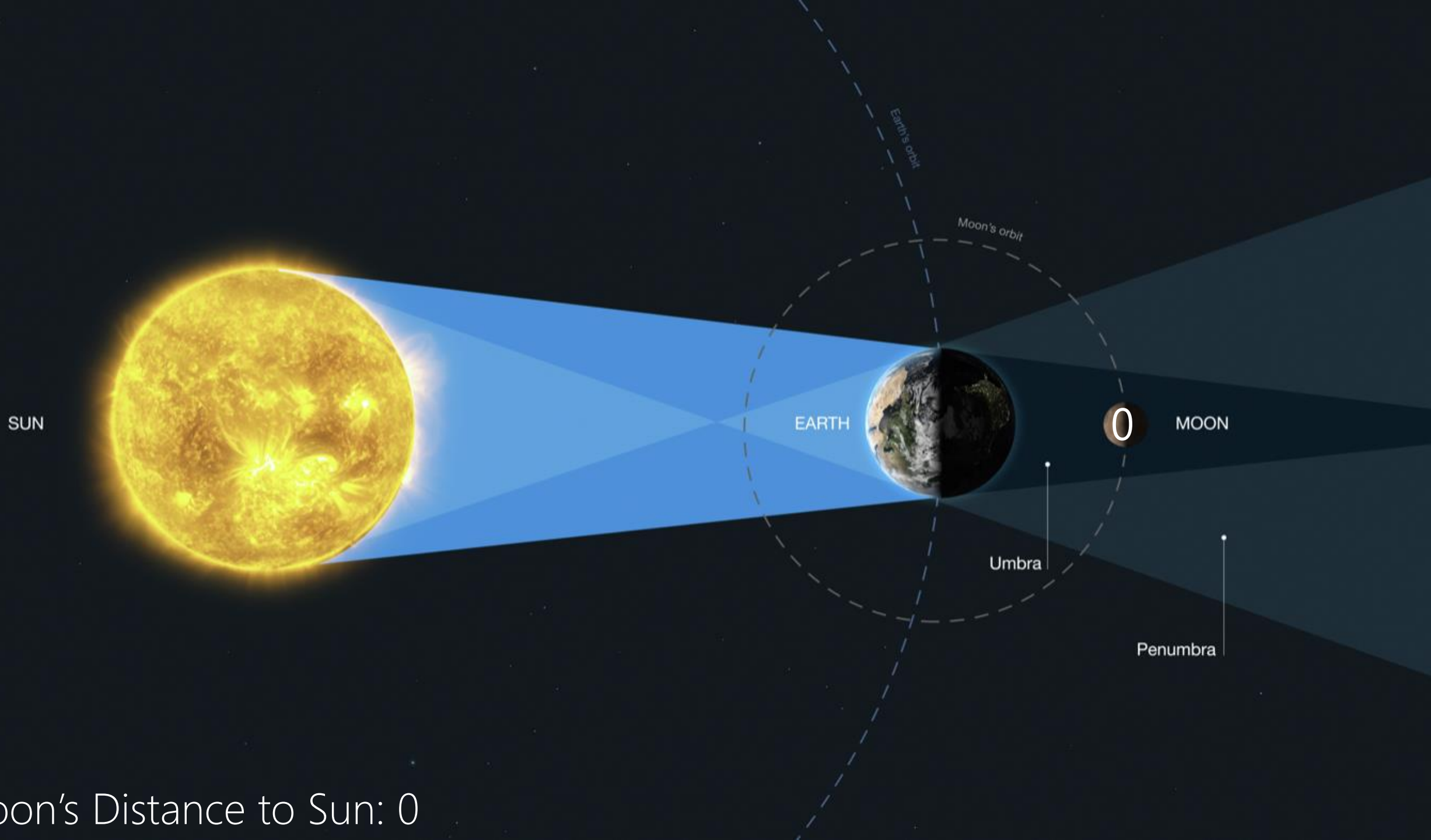
✂ Clip

🔖 Save

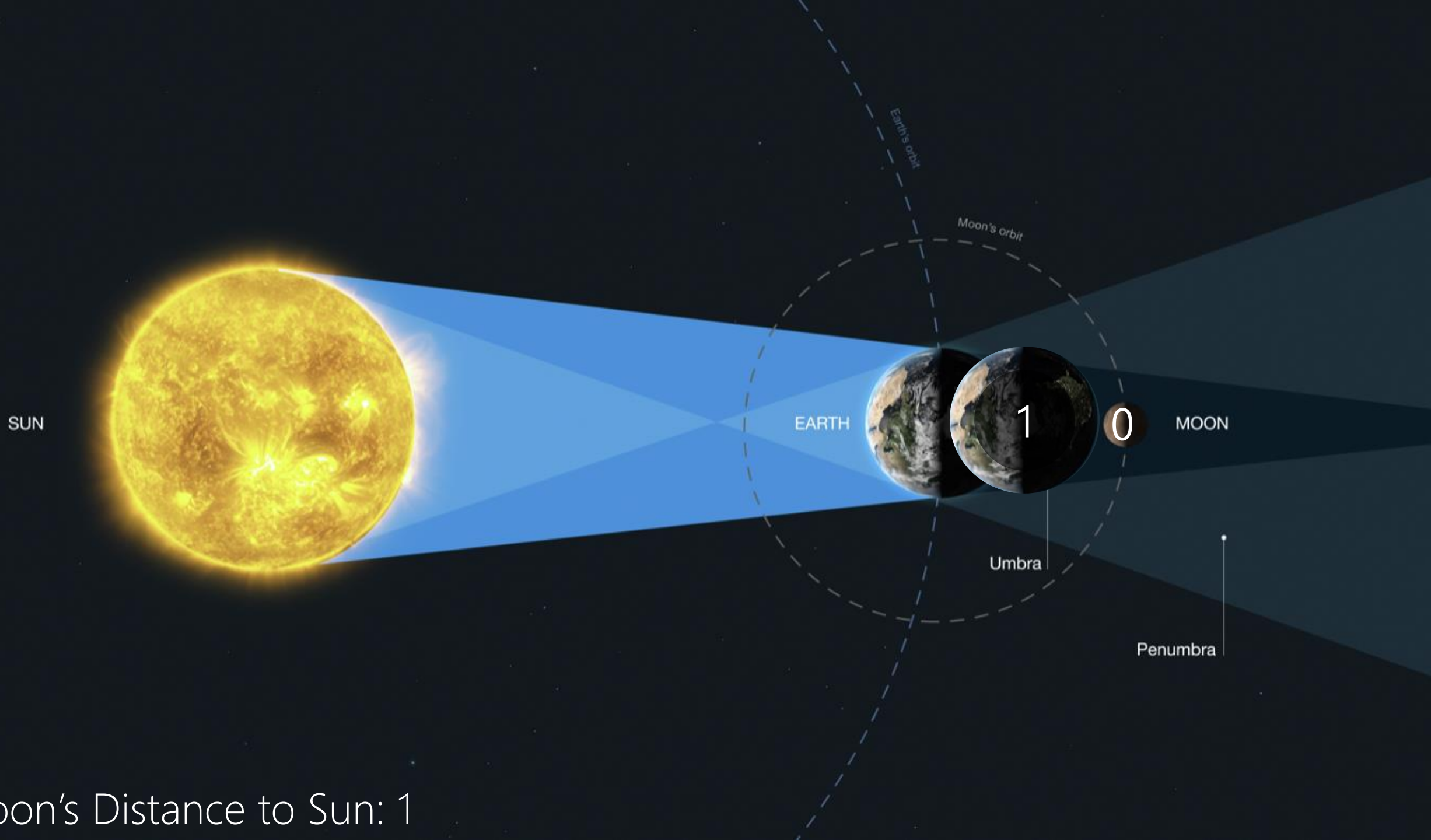


0123456789

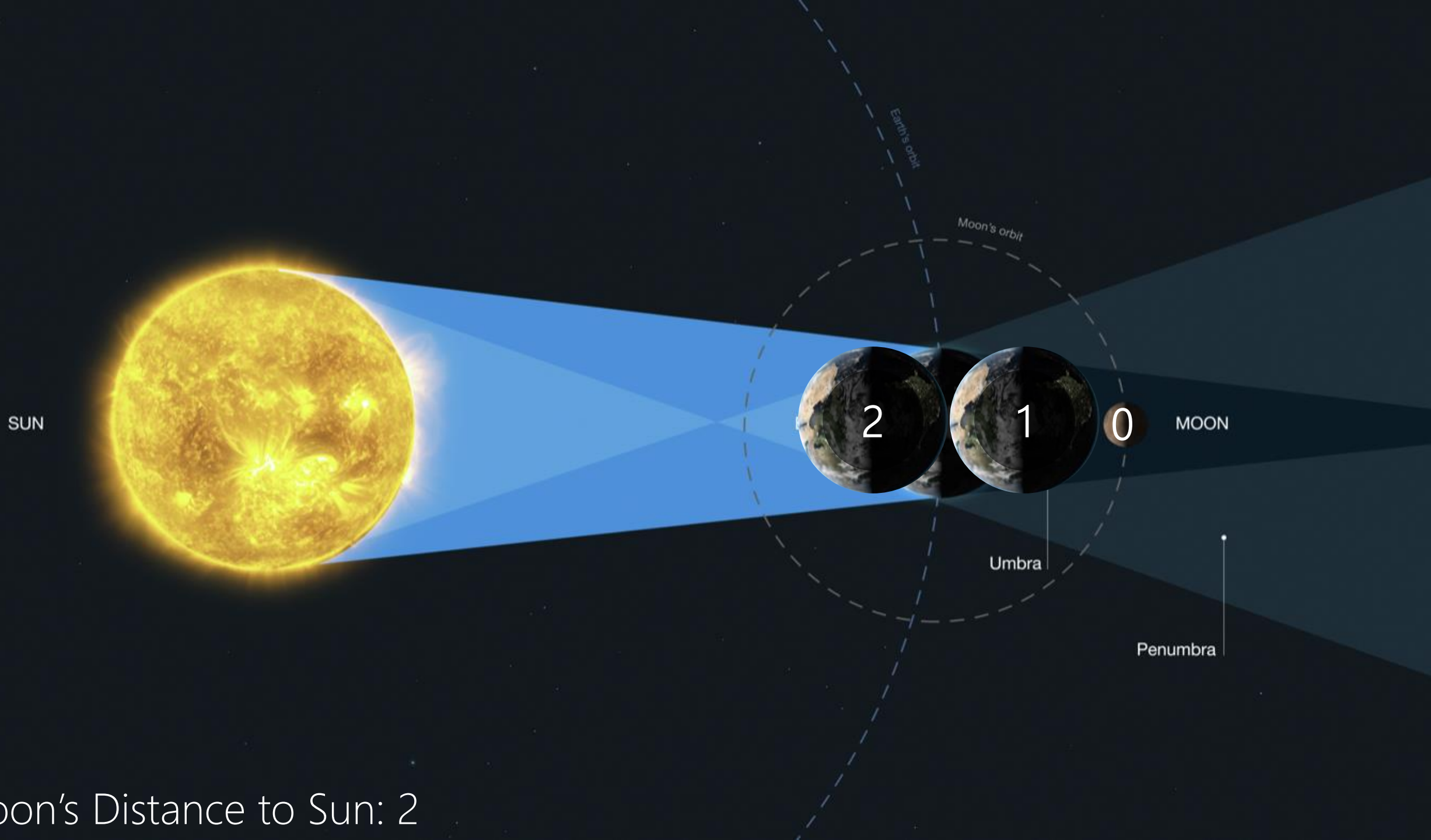
Hossein's Number System



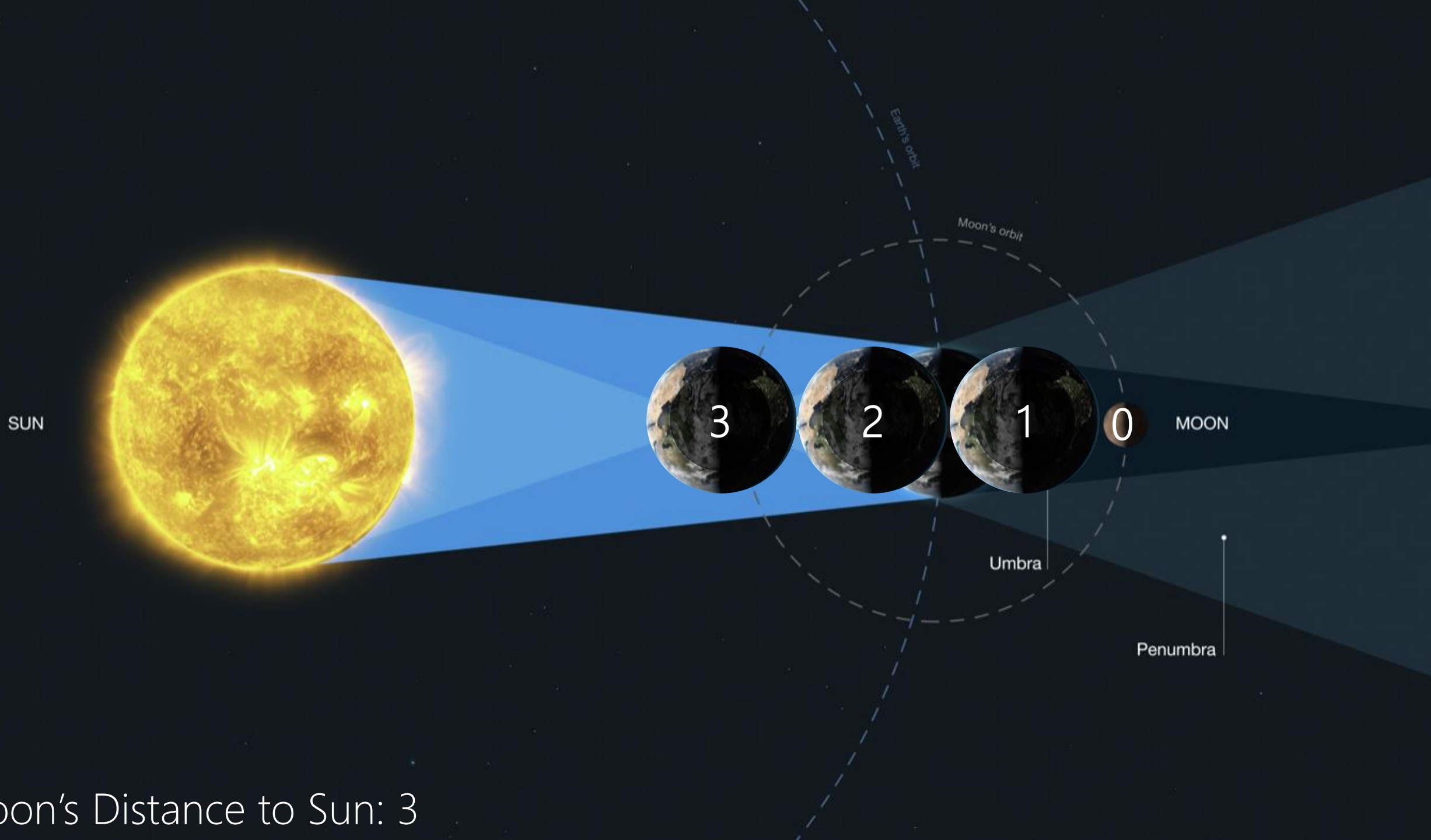
Moon's Distance to Sun: 0



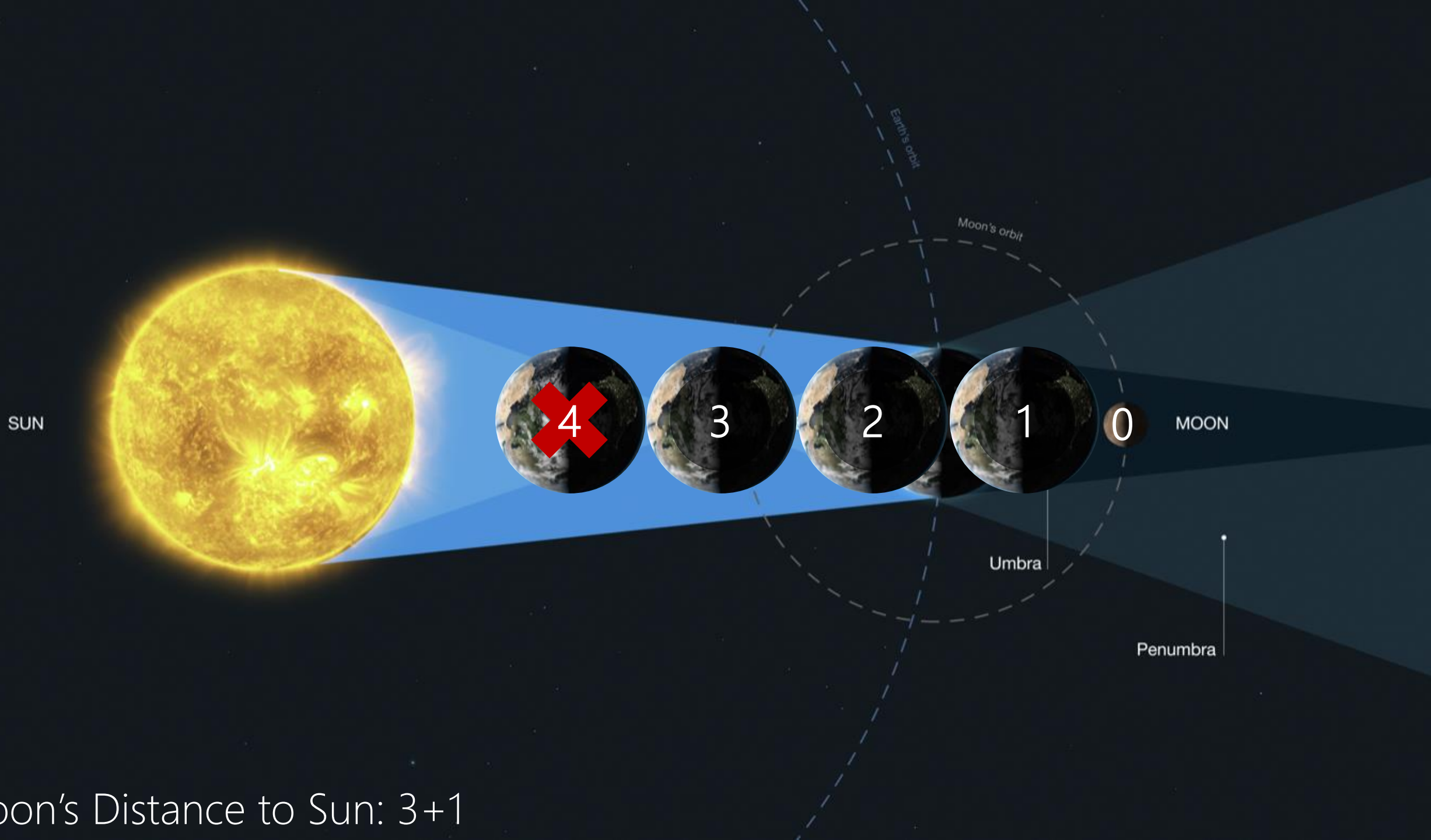
Moon's Distance to Sun: 1



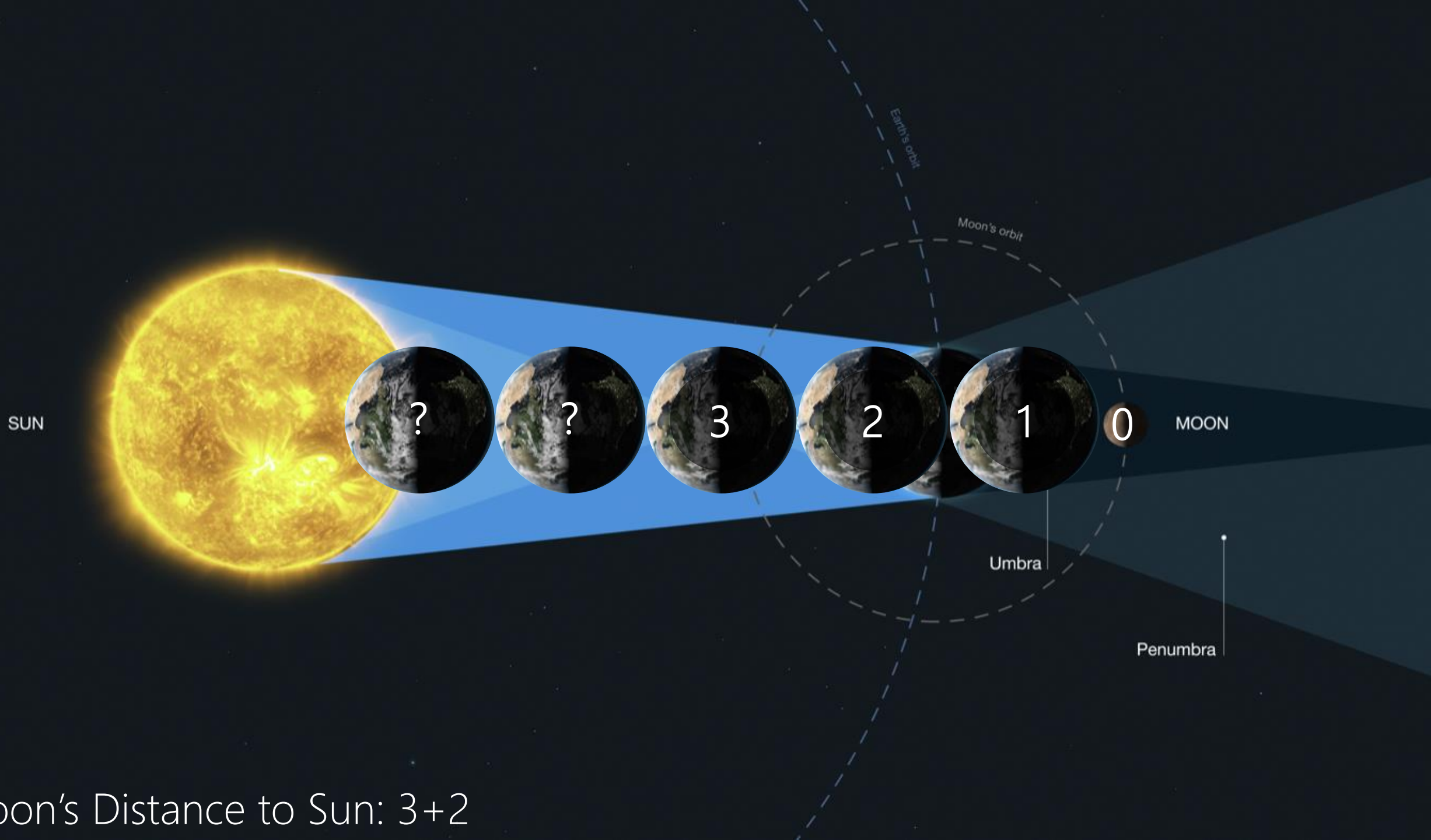
Moon's Distance to Sun: 2



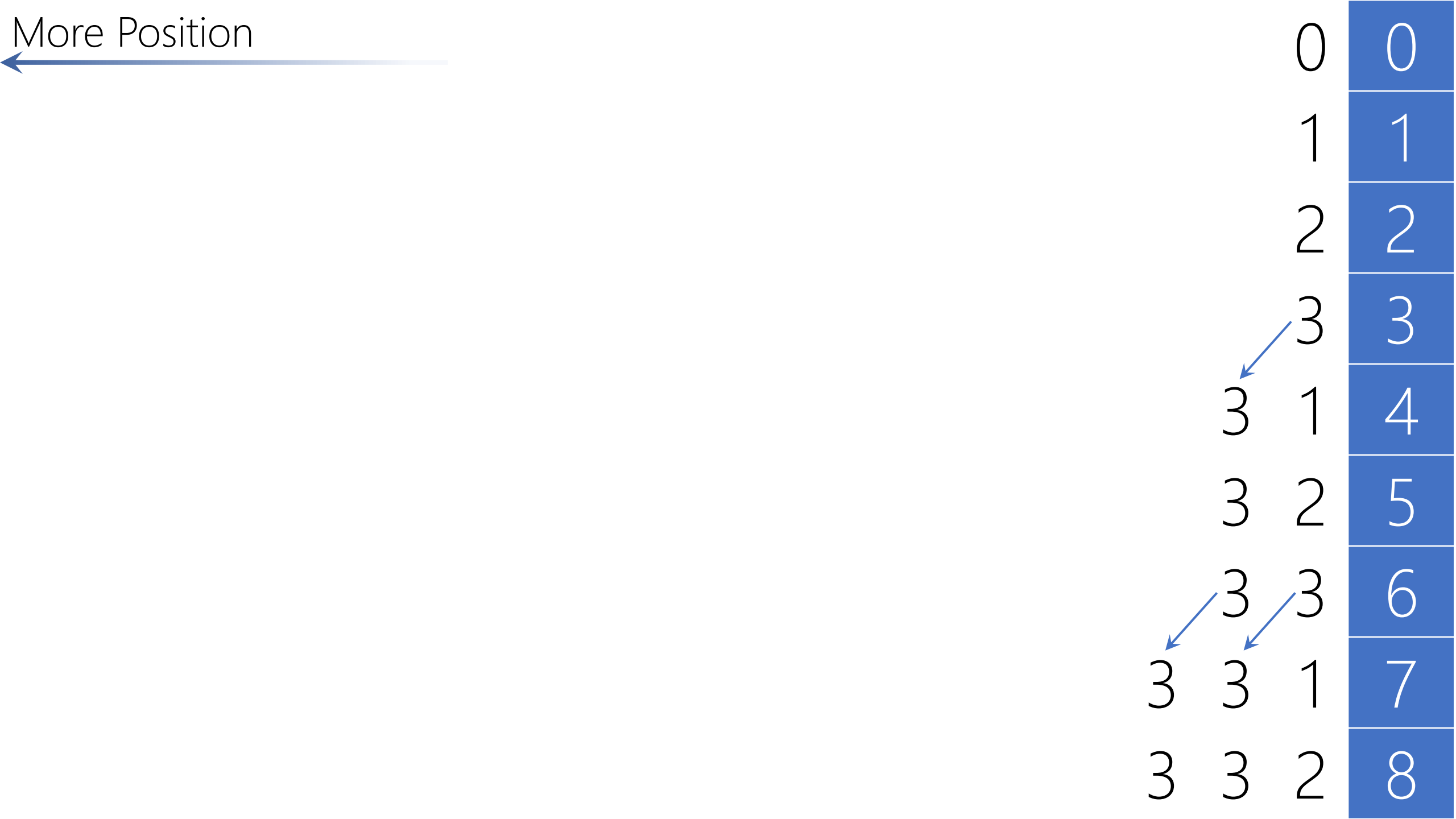
Moon's Distance to Sun: 3

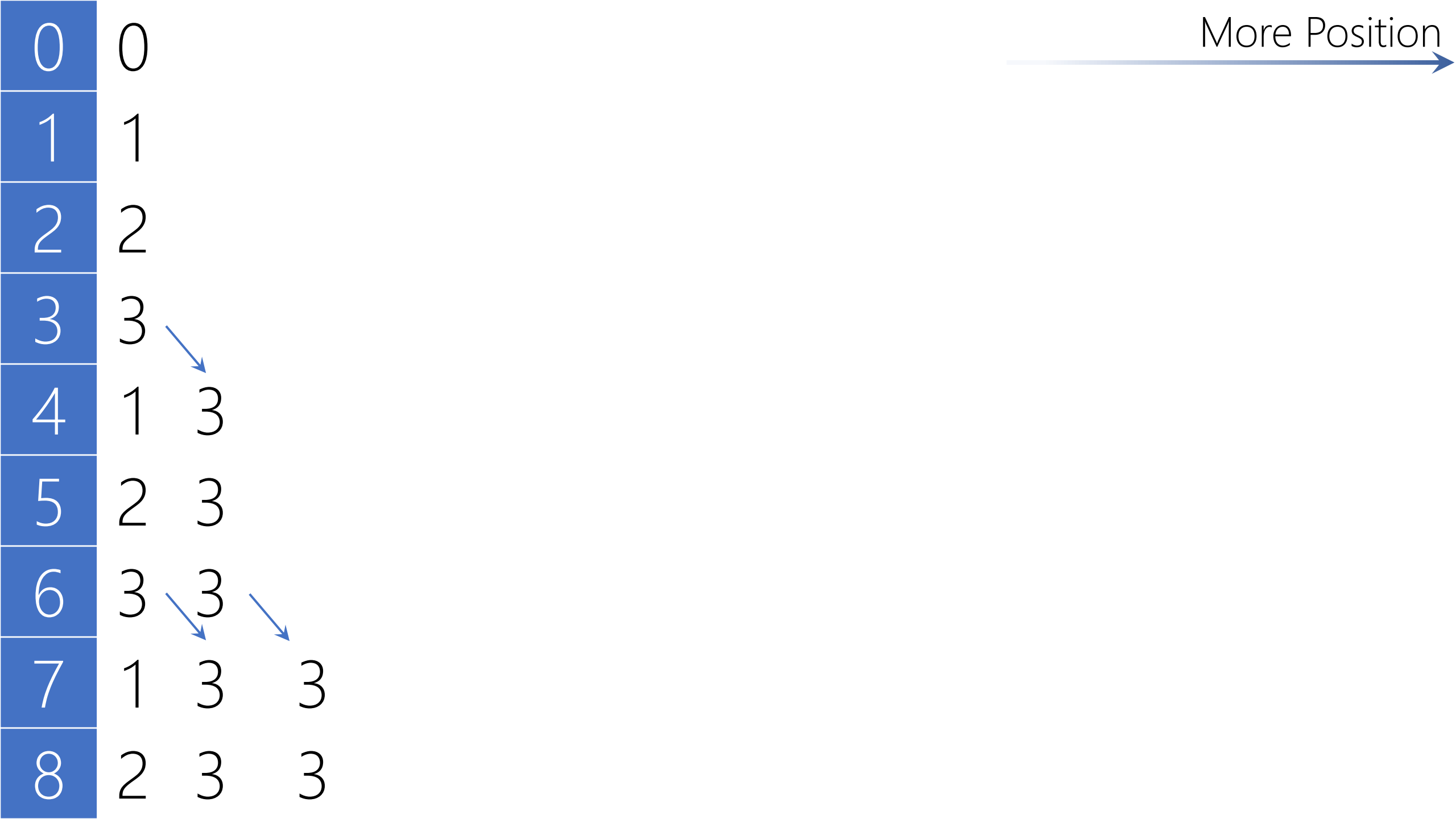


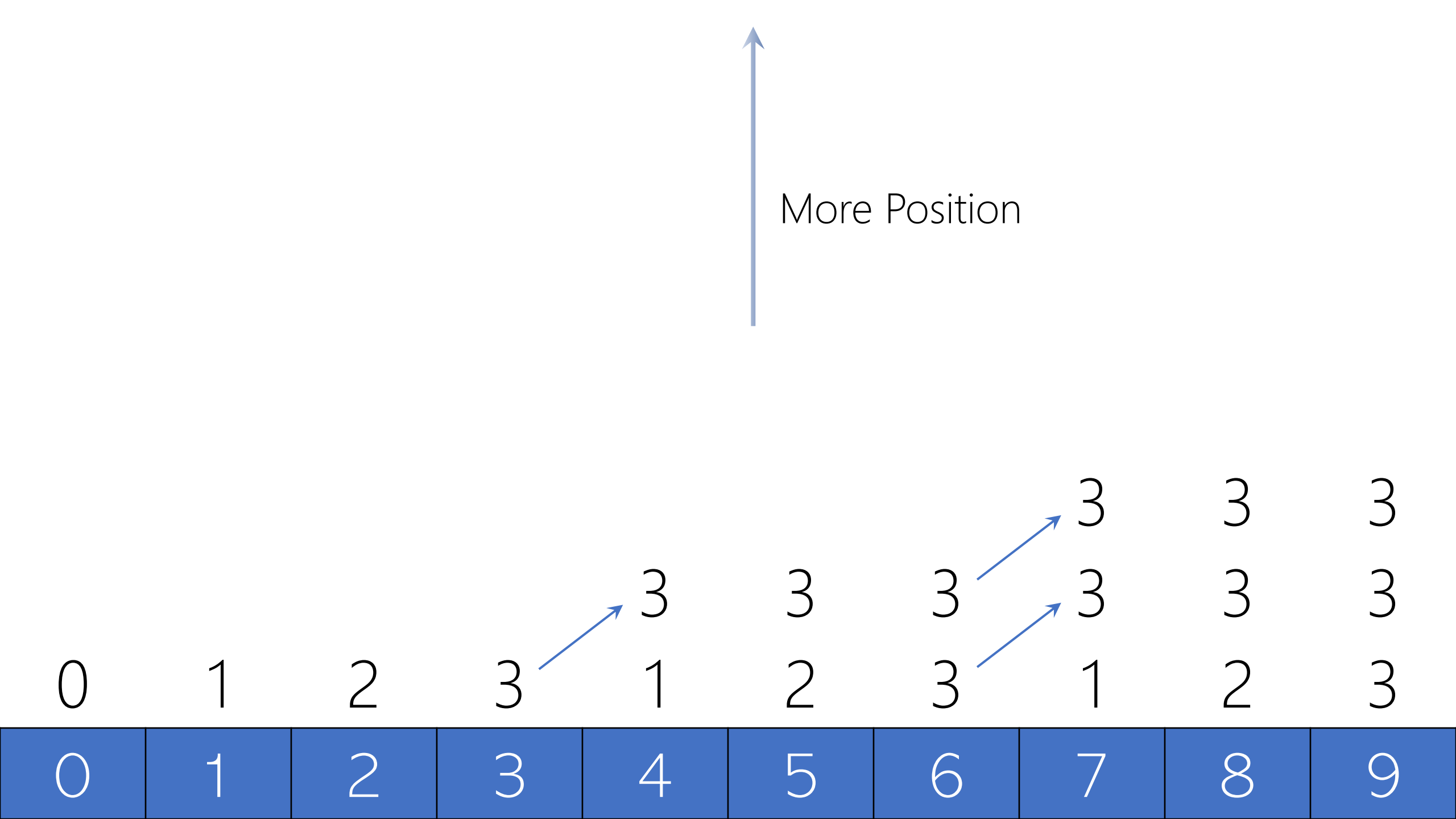
Moon's Distance to Sun: 3+1



Moon's Distance to Sun: $3+2$







More Position			0	$0=0$
			1	$1=1$
			2	$2=2$
			3	$3=3$
			3 1	$3+1=4$
			3 2	$3+2=5$
			3 3	$3+3=6$
			3 3 1	$3+3+1=7$
			3 3 2	$3+3+2=8$

[illegible]

[illegible]

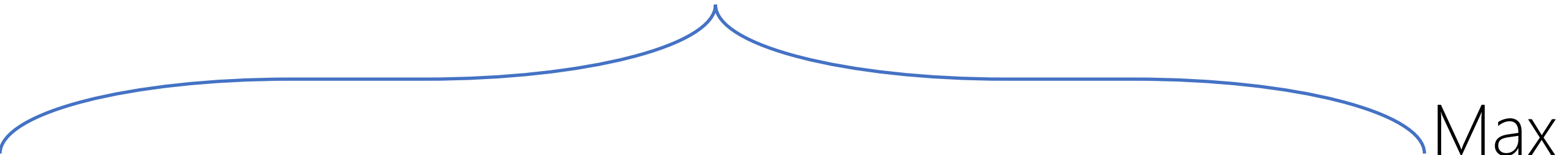
[illegible]

[illegible]

0	0	...			0	0	0
---	---	-----	--	--	---	---	---



n positions

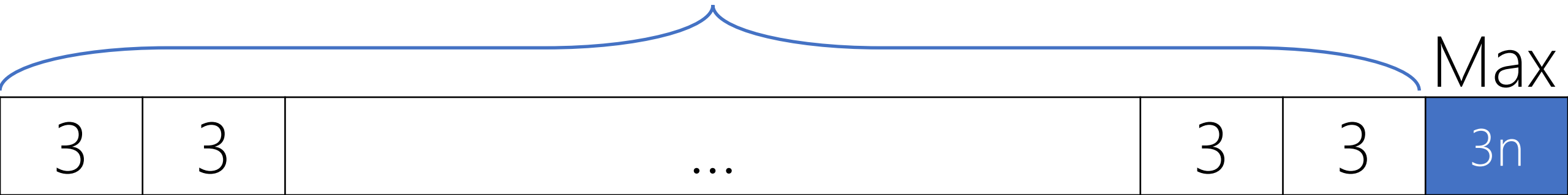


3	3	...			3	3	?
---	---	-----	--	--	---	---	---

What is the max number given n position?

$$3+3+3+\cdots+3 = 3 \times n = 3n$$

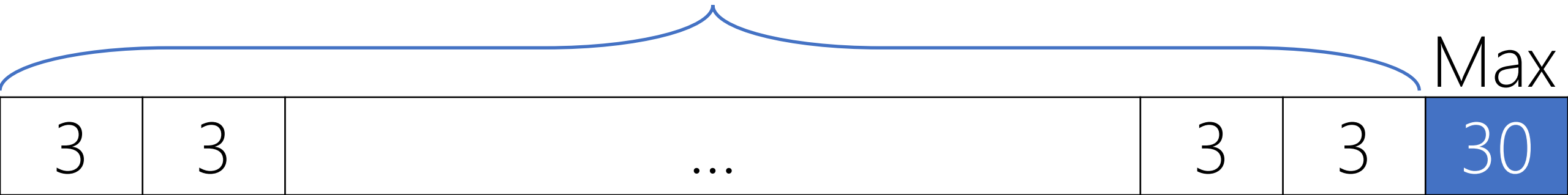
n positions



$$\text{Max} = 3n$$

$$n=10 \Rightarrow (3 \times 10) = 30$$

10 positions

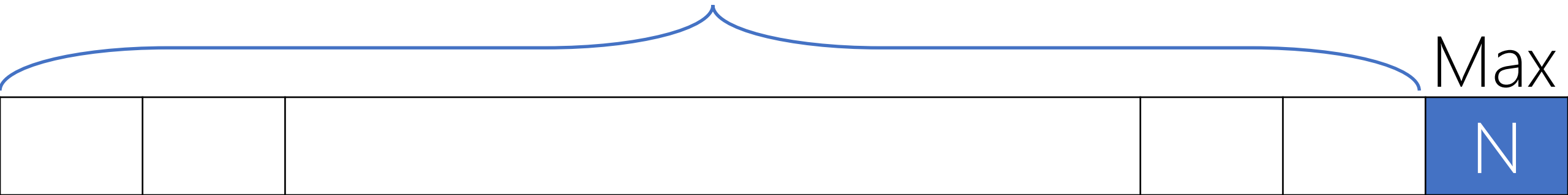


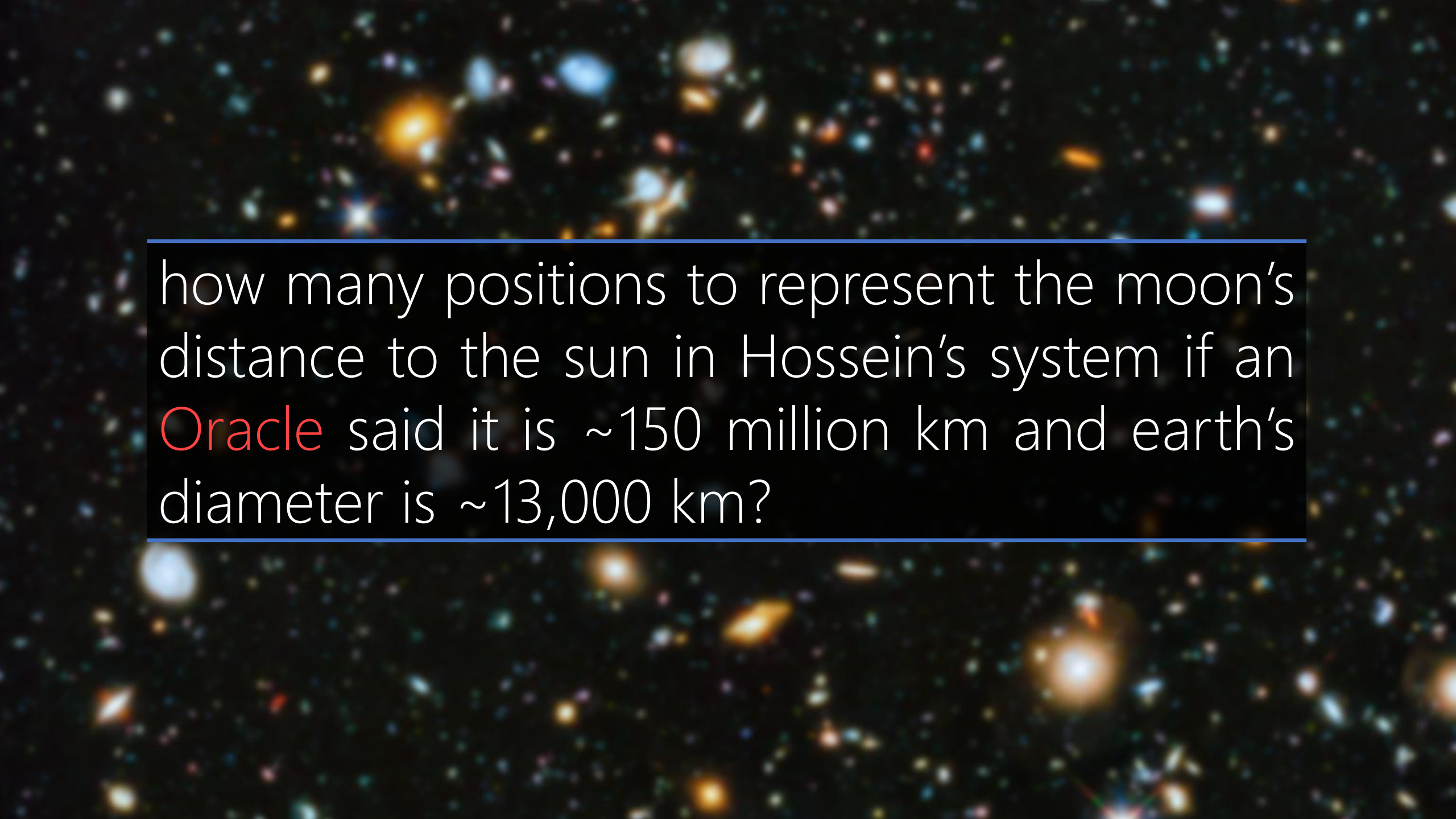
The other way around,
What is number of required position for a number N?

$$\text{Max} = N = 3n$$

$$N \div 3 = n$$

? positions





how many positions to represent the moon's distance to the sun in Hossein's system if an **Oracle** said it is ~150 million km and earth's diameter is ~13,000 km?

$\sim 150 \text{ million km} \div \sim 13,000 \text{ km} = \sim 12,000 \text{ Earth}$

$$N = 12,000$$

$$n = N \div 3 = 12,000 \div 3 = 4,000 \text{ positions}$$

paper = $\sim 3,000$ positions

$$4,000 \div 3,000 = 1.33 \text{ pages} = 2 \text{ pages}$$

